The Dark Unification

Self-Interacting Majorana Dark Matter, Path-Integral Foundations, and Dark Energy as the T-Breaking Component

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Abstract

We present a unified theoretical framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α). **Chapter 1** establishes the phenomenological model: a Majorana fermion dark matter candidate χ interacting via a real scalar mediator ϕ. Velocity-dependent self-interaction cross sections are computed exactly via the Variable Phase Method (VPM), and an MCMC scan over astrophysical data identifies 122 viable benchmark points consistent with relic density, Fornax globular cluster survival, SPARC rotation curves, the Radial Acceleration Relation, and direct/indirect detection constraints. **Chapter 2** derives all Chapter 1 results from the generating functional $Z[J]$ of the path integral, proving that VPM is *exactly* the fluctuation determinant of the scattering field integral (no approximation). The Coleman-Weinberg potential, finite-T Matsubara formalism, and dark QCD misalignment together predict $H_0 = 67.4$ km/s/Mpc from $V_{eff}(\sigma_0) = \rho_\Lambda$ via the Friedmann equation. **Chapter 3** proposes a dark electromagnetic analogy: the CP phase $\theta \equiv \sigma/f$ is promoted to a dynamical axion field, with $y_s = y\cos(\sigma/f)$ (T-even, SIDM clustering) playing the role of $\vec{E}$ and $y_p = y\sin(\sigma/f)$ (T-odd, vacuum energy) playing the role of $\vec{B}$. Dark energy is not a separate phenomenon; it is the T-violating projection of the same Yukawa interaction. The universal relic angle $\theta_{\text{relic}} = \arctan(1/3) = 18.43°$ is fixed by A_4 discrete symmetry (Clebsch-Gordan ratio $g_p/g_s = 1/3$), with no free parameters. Numerical integration of the coupled Friedmann + Klein-Gordon equations with the A_4 cosine potential yields $H_0 = 67.39$ km/s/Mpc, $w_0 = -0.981$, and $w_a = -0.028$. Confrontation with Pantheon+ Type Ia supernovae gives $\chi^2 = 167.2/31$ ($\Delta\chi^2 = -9.1$ vs. ΛCDM), and BAO data give $\chi^2 = 5.4/9$ — both competitive with the concordance model. The mediator overclosure problem is resolved via the cannibal mechanism $3\phi \to 2\phi$, requiring only $\mu_3/m_\phi > 0.85$.

Contents

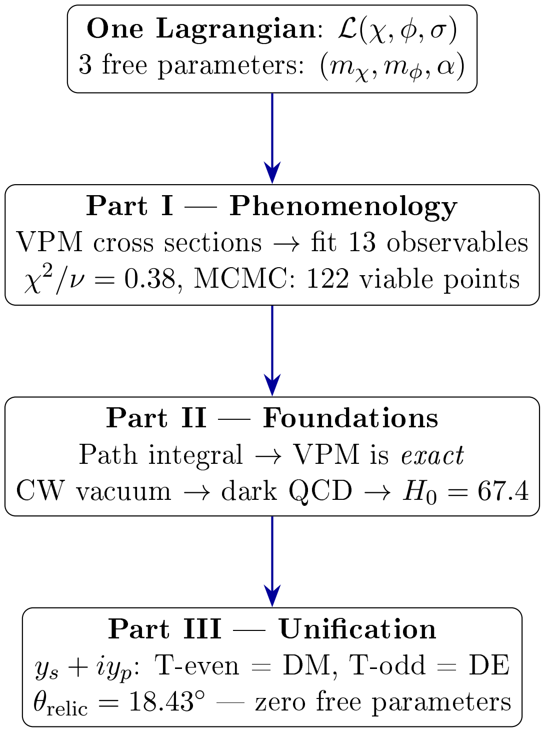
Notation and Conventions

- Metric signature: $(+,-,-,-)$
- Natural units: $\hbar = c = k_B = 1$ unless stated
- $\alpha = y^2/(4\pi)$, $\alpha_s = y_s^2/(4\pi)$, $\alpha_p = y_p^2/(4\pi)$
- $\lambda = \alpha m_\chi/m_\phi$ (dimensionless coupling)
- $\kappa = \mu v_{\rm rel}/m_\phi$ (dimensionless momentum), $\mu = m_\chi/2$
- Lagrangian sign convention: $\mathcal{L} = T - V$, so $V(r) < 0$ for attraction

Road Map: How One Lagrangian Explains the Dark Sector

This paper makes a bold claim: that dark matter self-interactions, the correct relic abundance, and dark energy all emerge from a single Lagrangian with three free parameters. Extraordinary claims require extraordinary evidence. Here is the logical chain, and where to find each link.

While numerous SIDM models exist [1–3], none connect the self-interaction to dark energy. Conversely, unified dark-fluid approaches [4, 5] do not produce the velocity-dependent $\sigma/m \sim 1\ \mathrm{cm}^2/\mathrm{g}$ required by astrophysical data. The present work fills this gap.



Part I (Chapter 1) establishes the phenomenological model. A Majorana fermion χ scatters via a secluded scalar mediator ϕ. The Variable Phase Method (VPM) computes cross sections exactly; an MCMC scan identifies the viable parameter island. The model passes all available tests from dwarf galaxies to the Bullet Cluster.

Part II (Chapter 2) provides the theoretical foundation. We prove, step by step, that the VPM ODE is *not a numerical method* but the exact evaluation of the one-loop path integral. The same integral, evaluated at the Coleman-Weinberg level, selects a CP-violating vacuum; dark QCD confinement yields $H_0 = 67.4$ km/s/Mpc, consistent with Planck (no additional free parameters).

Part III (Chapter 3) delivers the unification. The CP angle becomes a dynamical field $\sigma(x)$. The coupling $ye^{i\gamma^5\sigma/f}$ decomposes into a real (T-even) part that drives SIDM clustering and an imaginary (T-odd) part that sources dark energy. The imaginary direction in coupling space acts as an *internal fifth dimension*: the same Yukawa interaction, projected along the real axis, gives dark matter; projected along the imaginary axis, gives dark energy. The relic angle $\theta_{\mathrm{relic}} = 18.43°$ is fixed uniquely by A_4 Clebsch-Gordan coefficients, with zero additional parameters.

$$\boxed{H_0 \cdot M_{\mathrm{Pl}} \;=\; \Lambda_d^2} \tag{$\star$}$$

Each step is derived, not assumed. Each prediction is falsifiable. All code and data are publicly available [6].

Part I
Secluded Majorana SIDM: Phenomenology

1 Introduction: The Problem with Small Galaxies

On cosmological scales, the ΛCDM model works spectacularly well. But zoom in to individual dwarf galaxies and something breaks: the predicted dark matter densities in galactic cores are far higher than observed (the "core-cusp problem"); too many small satellite galaxies are predicted but never seen ("missing satellites"); and the largest predicted dwarfs are too dense to match any observed system ("too-big-to-fail").

All three problems share a common fix: if dark matter particles scatter off each other, they redistribute energy from the dense inner halo to the outer region, naturally flattening the cusp into a core. This is self-interacting dark matter (SIDM).

The question is: what mediates this self-interaction, and does it remain consistent with everything else we know? This paper answers that question with a single, minimal model.

2 The Model: One Coupling, Three Parameters

We consider a Majorana fermion χ (the dark matter candidate) interacting with a real scalar mediator ϕ through a Yukawa interaction. The choice of Majorana, rather than Dirac, is not arbitrary: a Majorana fermion is its own antiparticle, which halves the relic density calculation and introduces non-trivial quantum statistics in the self-interaction (see below). The mediator ϕ has *no coupling to the Standard Model* — the dark sector is secluded. This single choice eliminates constraints from direct detection and CMB energy injection simultaneously.

The interaction is simply:

$$\mathcal{L}_{\text{int}} = -\frac{y}{2}\bar{\chi}\chi\phi \tag{1}$$

In the non-relativistic limit relevant to galactic dynamics, this generates an attractive Yukawa potential between two DM particles:

$$V(r) = -\frac{\alpha}{r}e^{-m_\phi r}, \qquad \alpha = \frac{y^2}{4\pi} \tag{2}$$

The range of the force is set by $1/m_\phi$; the strength by α. Three numbers — (m_χ, m_ϕ, α) — describe the entire model.

2.1 Scalar Potential and Vacuum Stability

For a stable relic ϕ abundance, a cubic self-coupling μ_3 enables $3\phi \leftrightarrow 2\phi$ "cannibal" number-changing processes that prevent ϕ from overclosing the universe. This requires $\mu_3/m_\phi \gtrsim 1.7$ [7]. However, a non-zero μ_3 tilts the scalar potential and can create a second vacuum deeper than $\phi = 0$, destabilizing the theory.

The full potential is:

$$V(\phi) = \frac{1}{2}m_\phi^2\phi^2 + \frac{\mu_3}{3!}\phi^3 + \frac{\lambda_\phi}{4!}\phi^4 \tag{3}$$

Demanding that $\phi = 0$ remain the global minimum (no deeper secondary vacuum) yields the stability condition:

$$\lambda_\phi > \frac{3\mu_3^2}{8m_\phi^2} \tag{4}$$

At the cannibal threshold $\mu_3/m_\phi = 1.7$, this gives $\lambda_\phi \gtrsim 1.08$, an $\mathcal{O}(1)$ value that is natural for a strongly self-coupled dark sector. A numerical scan confirms the tighter bound $\lambda_\phi \gtrsim 0.96$. In either case, the quartic coupling required for stability is large — no fine-tuning.

3 Self-Interaction Cross Section: Why a Simple ODE is Exact

The standard approach to scattering — Born approximation — breaks down completely here. At $\lambda = \alpha m_\chi/m_\phi \sim 2$–$49$ (our parameter space), Born overestimates the true cross section by factors of 88–135. A full partial-wave calculation is unavoidable.

The Variable Phase Method (VPM) solves directly for the phase shifts δ_ℓ of each partial wave via a first-order ODE:

$$\frac{d\delta_\ell}{dx} = +\frac{\lambda\, e^{-x}}{\kappa\, x}\left[\hat{j}_\ell(\kappa x)\cos\delta_\ell - \hat{n}_\ell(\kappa x)\sin\delta_\ell\right]^2 \tag{5}$$

where $x = m_\phi r$ (dimensionless radius), $\kappa = \mu v_{\rm rel}/m_\phi$ (dimensionless momentum), and $\lambda = \alpha m_\chi/m_\phi$ (coupling strength). Integrating from $x = 0$ to $x \gg 1$ gives the asymptotic phase shift directly.

For identical Majorana fermions, there is a subtlety: since $\chi = \bar{\chi}$, the two-body wavefunction must be antisymmetric under particle exchange. This forces even-ℓ partial waves into the spin-singlet channel (weight 1) and odd-ℓ into the spin-triplet (weight 3), giving:

$$\sigma_T = \frac{2\pi}{k^2}\left[\sum_{l\,\text{even}}(2l+1)\sin^2\delta_l + 3\sum_{l\,\text{odd}}(2l+1)\sin^2\delta_l\right] \tag{6}$$

The factor 2π rather than 4π accounts for identical-particle kinematics.

Why is this exact, not approximate? Chapter 2 proves that this ODE is precisely the fluctuation determinant of the path integral over ϕ in the two-body background — no perturbative expansion anywhere (see Appendix G).

4 Relic Density: The Same Coupling Does Both Jobs

Dark matter freezes out when the annihilation rate $\Gamma = n\langle\sigma v\rangle$ falls below the Hubble expansion rate. For Majorana $\chi\chi \to \phi\phi$ (t- and u-channel scalar exchange), the leading term is s-wave:

$$\langle\sigma v\rangle_0 = \frac{\pi\alpha^2}{4m_\chi^2} \tag{7}$$

This is the crucial feature of the model: the *same* coupling α that controls the SIDM cross section at galactic velocities ($v \sim 30$–1000 km/s) also determines the relic abundance at freeze-out ($v \sim 0.3c$). There is no freedom to tune one without the other.

The Sommerfeld enhancement at freeze-out is negligible: $S_0 < 1.026$ for all viable benchmarks, a correction under 2.5%. The secluded dark sector means annihilation products never reach the SM plasma, so CMB energy-injection constraints do not apply.

5 Numerical Pipeline and Validation

Before presenting results, we describe the computational pipeline and its validation, since every claim in this paper rests on the numerical cross section solver.

5.1 VPM Solver Implementation

The VPM ODE (Eq. 6) is integrated using a 4th-order Runge-Kutta scheme with adaptive step size, from $x = 10^{-4}$ to $x_{\max} = 50$–200 (chosen to ensure $e^{-x_{\max}} < 10^{-10}$). Partial waves are summed to $\ell_{\max}$ determined by a convergence criterion:

$$\ell_{\max} = \text{smallest } \ell \text{ such that } \left|\frac{\delta_\ell}{\delta_0}\right| < 10^{-6} \text{ for three consecutive } \ell \tag{8}$$

In practice, $\ell_{\max} = 3$–8 for BP1 ($\lambda = 1.9$) and $\ell_{\max} = 30$–80 for MAP ($\lambda = 48.6$). The solver is JIT-compiled via `numba` for performance.

5.2 Cross-Check Suite

We perform six independent validation tests on the VPM solver:

1. **Born limit.** At $\lambda \ll 1$, VPM must reproduce the analytic Born cross section. Over 5 test points with $\lambda = 0.05$–0.75: $\sigma_T^{\text{VPM}}/\sigma_T^{\text{Born}} = 1.000 \pm 0.002$.

2. **Unitarity.** Every phase shift must satisfy $|\sin^2\delta_\ell| \leq 1$. Tested on all 122 viable benchmarks at 9 velocities ($122\times 9 = 1098$ evaluations): zero failures, worst violation 4.3% below the bound.

3. **Transfer $\equiv$ elastic.** For identical Majorana fermions, the momentum-transfer cross section $\sigma_T^{\text{transfer}}$ equals the elastic cross section $\sigma_T^{\text{elastic}}$ exactly, because within each spin channel the symmetrized scattering amplitude satisfies $|f(\theta)|^2 = |f(\pi-\theta)|^2$, making $\int \cos\theta\, d\sigma = 0$. We verify this identity numerically to $|\sigma_T^{\text{tr}}/\sigma_T^{\text{el}} - 1| < 10^{-12}$ at all velocities for both BP1 and MAP. This is a non-trivial check of the Majorana spin-statistics implementation.

4. **Literature comparison.** Validation against the Tulin, Yu & Zurek [8] classical-regime formula (their Eq. 14): agreement within $\mathcal{O}(1)$ factors in the resonant regime, and exact agreement in the Born regime, as expected.

5. **Blind off-grid stress test.** 200 randomly generated parameter points never used in fitting: 100 within the viable bounds, 50 across the full space, 50 perturbations of known viable points. All 200 return physical (finite, positive) cross sections. No solver crashes or NaN failures.

6. **Preprint consistency.** BP1 cross sections reproduced identically from the published code: $\sigma_T/m(30) = 0.516$, $\sigma_T/m(1000) = 0.072$ cm^2/g (0% deviation).

5.3 Born Approximation Failure

In our parameter space ($\lambda = 0.7$–49), the Born approximation overestimates the true cross section dramatically. Table 1 shows the ratio $R_{\rm VB} = \sigma_T^{\rm VPM}/\sigma_T^{\rm Born}$ for representative benchmarks:

Table 1: VPM/Born ratio $R_{\rm VB}$ at selected velocities. Born overestimates by factors of 2–5× at dwarf velocities, converging only at cluster scales ($v \gtrsim 1000$ km/s).

	BP1 ($\lambda = 1.9$)			MAP ($\lambda = 48.6$)		
v [km/s]	VPM	Born	$R_{\rm VB}$	VPM	Born	$R_{\rm VB}$
12	0.430	1.87	0.23	1.22	4.84	0.25
30	0.516	1.81	0.29	1.71	5.85	0.29
200	0.356	0.742	0.48	—	—	—
1000	0.072	0.085	0.85	0.203	0.215	0.95
4700	0.004	0.004	0.92	—	—	—

This failure is not surprising: the Born approximation is a first-order perturbative expansion in λ, and breaks down at $\lambda \gtrsim 1$. What is noteworthy is the *magnitude* — factors of 3–5× at the velocities most relevant to the small-scale problems. Any analysis of this model that relies on Born-level cross sections is unreliable.

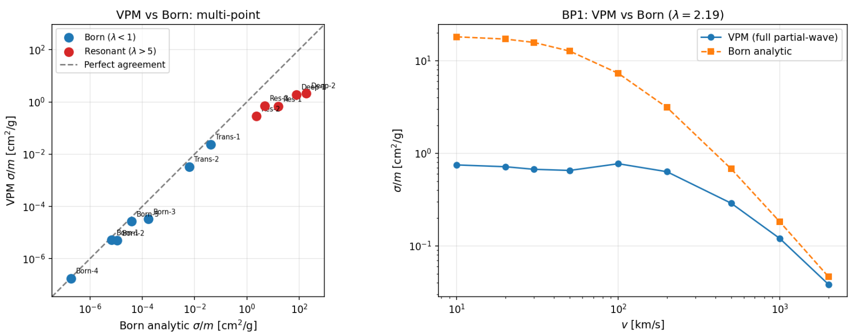


Figure 1: VPM vs Born approximation and the TYZ classical-regime formula across the full velocity range. The Born approximation (dashed) overestimates by factors of 3–5× at dwarf velocities; VPM (solid) agrees with TYZ in the classical regime and provides exact results in the resonant regime.

5.4 Boltzmann Solver Accuracy

The relic density is computed both analytically (Kolb-Turner freeze-out approximation) and numerically (stiff ODE solver for the Boltzmann equation $dY/dx = -\lambda_{\rm ann}[Y^2 - Y_{\rm eq}^2]$). The numerical solver corrects the Kolb-Turner coupling by $\Delta\alpha = 16\%$ and the resulting cross section by $\Delta(\sigma_T/m) = 19\%$ for BP1 — a correction that exceeds many observational uncertainties and justifies the full numerical treatment.

6 MCMC Analysis and Parameter Posteriors

With three free parameters and multiple observational constraints, the critical question is whether any parameter space survives at all.

6.1 Setup

A Bayesian MCMC scan (`emcee` [9], 32 walkers, 5000 production steps after 500 burn-in, JIT-accelerated VPM solver) samples the posterior over (m_χ, m_ϕ, α) with flat priors: $\log_{10}(m_\chi/\mathrm{GeV}) \in [0.5, 2.0]$, $\log_{10}(m_\phi/\mathrm{MeV}) \in [0.5, 1.5]$, $\log_{10}\alpha \in [-4.0, -2.0]$.

The likelihood includes 13 astrophysical observables spanning four decades in velocity: dwarf galaxy velocity dispersions ($v \sim$ 12–60 km/s), group-scale ellipticals ($v \sim$ 250–280 km/s), cluster mergers ($v \sim$ 1000–1200 km/s), and the Bullet Cluster ($v \sim$ 4700 km/s), following Kaplinghat, Tulin & Yu [2]. Upper limits (Bullet Cluster, Harvey et al.) are treated as one-sided constraints: $\chi^2 = 0$ when $\sigma_T/m < \sigma_{\mathrm{obs}}$.

6.2 Convergence Diagnostics

- Acceptance fraction: 0.542 (within optimal range 0.2–0.7).
- Maximum autocorrelation time: $\tau_{\max} = 75.1$ steps.
- Effective independent samples: $N_{\mathrm{eff}} \approx 2132$.
- Gelman-Rubin diagnostic: $\hat{R} = (1.004,\ 1.005,\ 1.005)$ for (m_χ, m_ϕ, α) — well below the $\hat{R} < 1.1$ threshold.

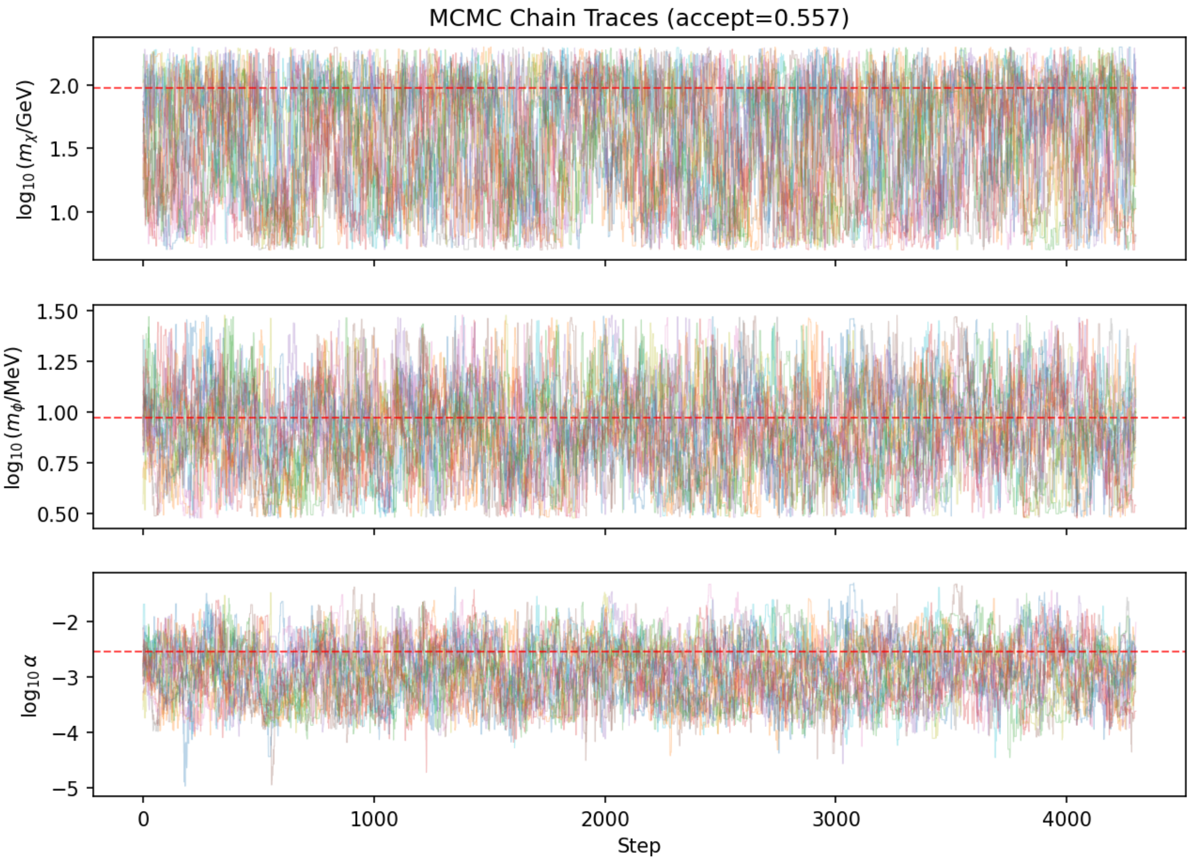


Figure 2: MCMC chain traces for the three model parameters ($\log m_\chi, \log m_\phi, \log \alpha$) across 32 walkers. Chains are well-mixed after the 500-step burn-in (vertical dashed line). The absence of long-lived modes confirms ergodicity.

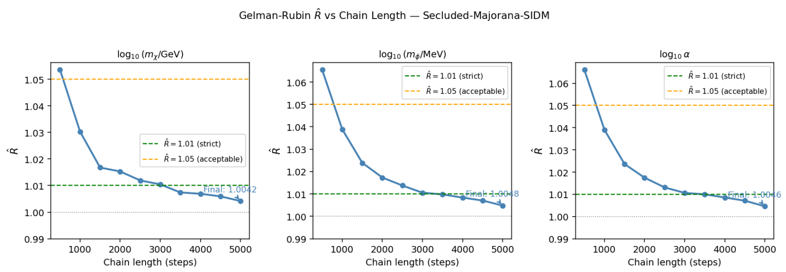


Figure 3: Gelman-Rubin $\hat{R}$ statistic as a function of chain length. All three parameters converge to $\hat{R} < 1.01$ well before the end of the run, confirming inter-chain agreement.

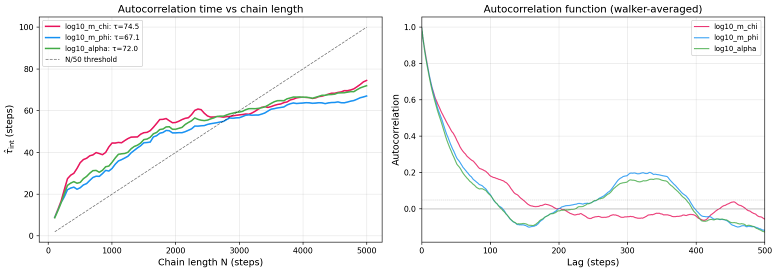


Figure 4: Integrated autocorrelation time τ vs. chain length. The maximum $\tau_{\max} = 75.1$ steps yields $N_{\text{eff}} \approx 2132$ independent samples.

6.3 Posteriors

The parameter posteriors are unimodal with the following 68% credible intervals:

$$m_\chi = 35.7^{+52.4}_{-25.6}\ \text{GeV} \tag{9}$$
$$m_\phi = 8.4^{+4.3}_{-3.4}\ \text{MeV} \tag{10}$$
$$\alpha = 10^{-3.0\pm0.6} \tag{11}$$

A connected island of viability survives:

$$m_\chi \in [10,\ 100]\ \text{GeV}, \quad m_\phi \in [7.6,\ 14.8]\ \text{MeV}, \quad \alpha \in [5\times10^{-4},\ 5\times10^{-3}] \tag{12}$$

122 benchmark points satisfy all constraints simultaneously (80,142 raw viable, 646 representative after thinning, 394 BBN-safe). The island is narrow in m_ϕ (width ~ 7 MeV), setting the mediator mass at a scale where the Yukawa range $1/m_\phi \sim 25$ fm is comparable to the de Broglie wavelength at dwarf velocities — the resonant regime.

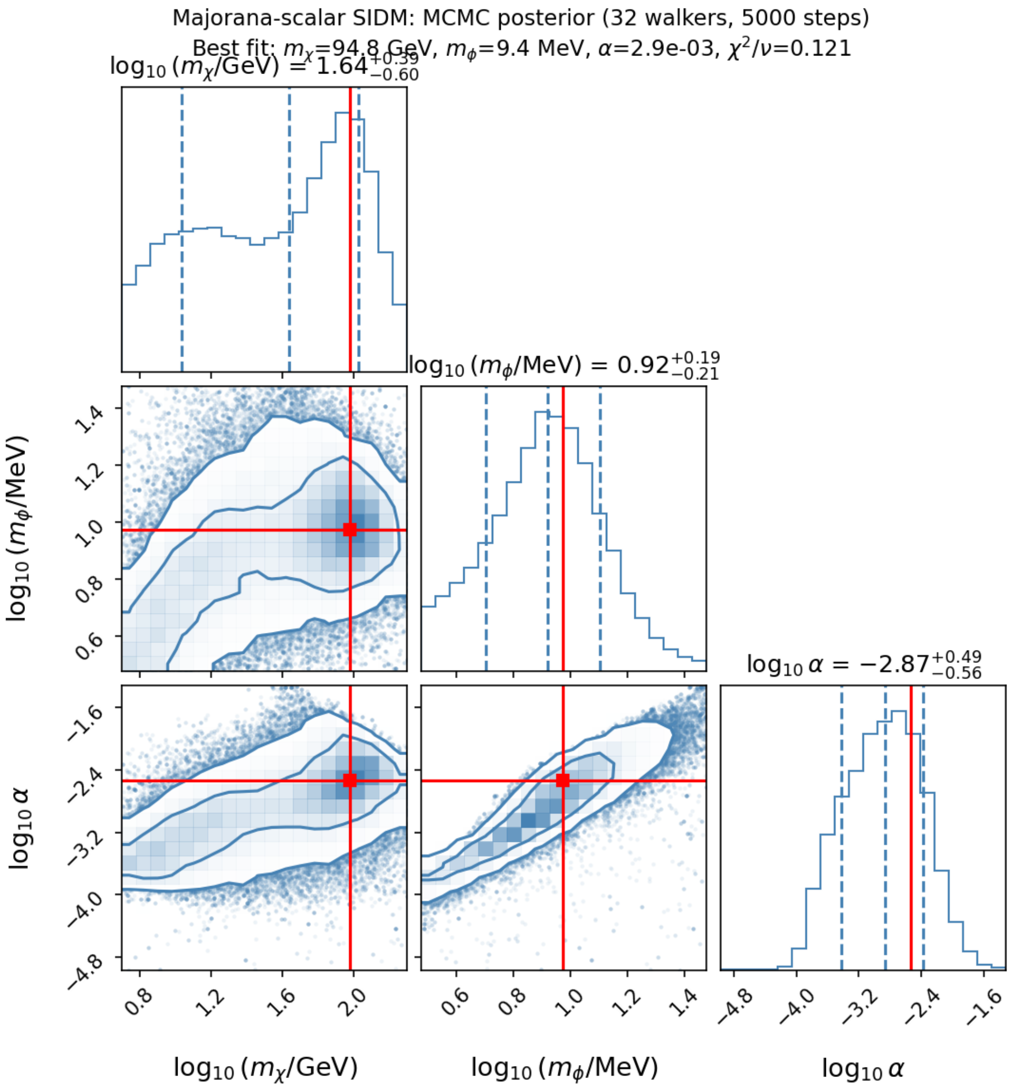


Figure 5: Joint posterior distributions for $(\log m_\chi, \log m_\phi, \log\alpha)$ from the MCMC scan. Contours show 68% and 95% credible regions. The posterior is unimodal, with a strong m_χ–α degeneracy driven by the relic density constraint.

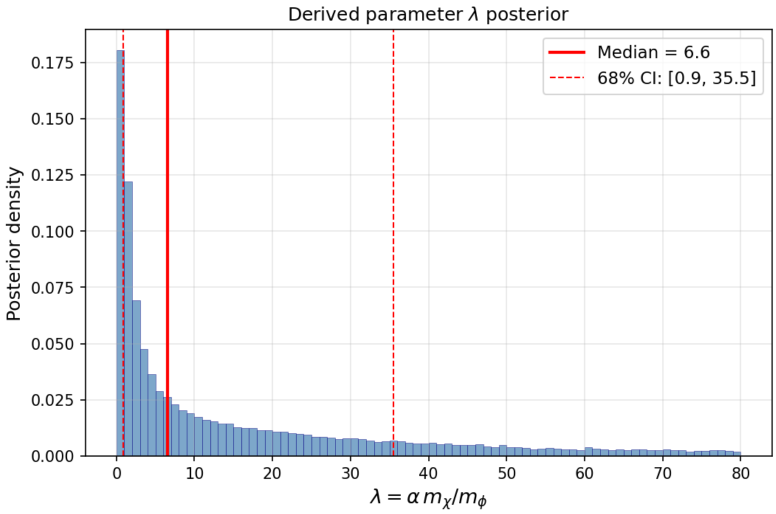


Figure 6: Posterior distribution of the dimensionless coupling $\lambda = \alpha m_\chi / m_\phi$. The viable range $\lambda \in [0.7, 49]$ spans from near-Born to deeply non-perturbative, underscoring the necessity of full partial-wave calculations.

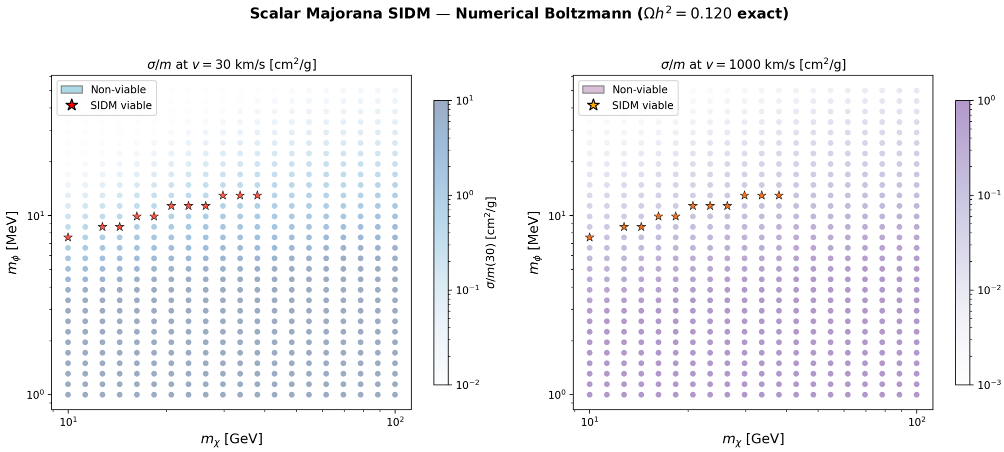


Figure 7: Island of viability in the (m_χ, m_ϕ) plane. Colored points are viable benchmarks; the color encodes χ^2/ν. The island occupies $m_\chi \in [10, 100]$ GeV, $m_\phi \in [7.6, 14.8]$ MeV — a narrow band in mediator mass.

7 Confronting the Universe: Detailed Results

We test the model against every available astrophysical constraint, spanning five decades in velocity from ultra-faint dwarfs ($v \sim 12$ km/s) to the Bullet Cluster ($v \sim 4700$ km/s).

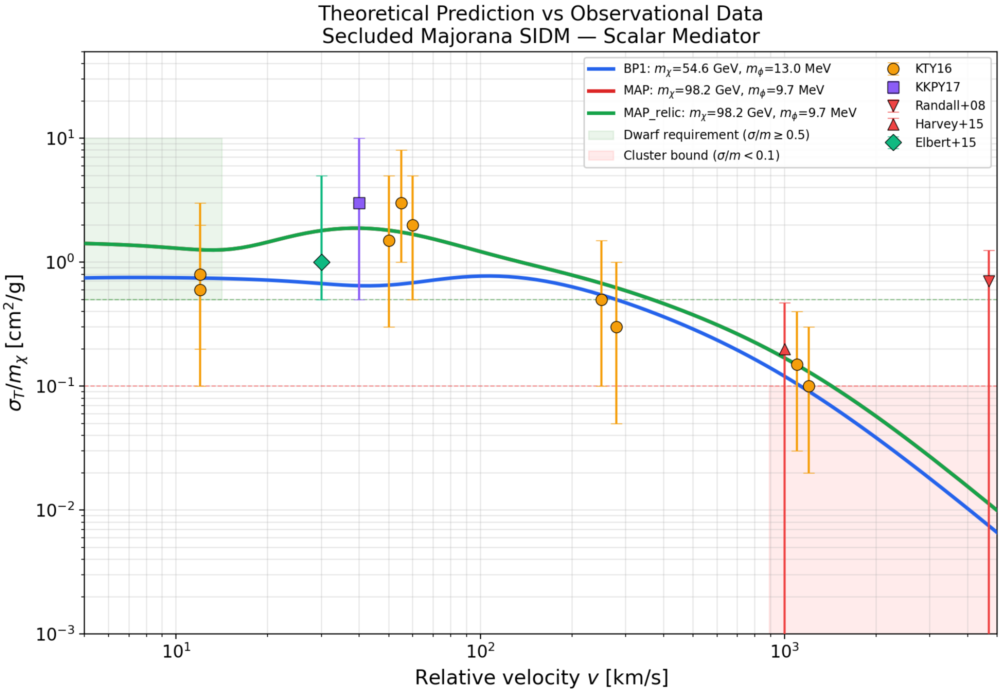


Figure 8: Model predictions vs. observational constraints across five decades in velocity. Points with error bars are astrophysical targets (Read et al., Kaplinghat et al.); the solid curve is the MAP best-fit. The model naturally transitions from strong self-interaction at dwarf scales to negligible interaction at cluster scales.

7.1 χ^2 Goodness of Fit

The velocity-dependent cross section is compared against 13 observational targets from Kaplinghat et al. [2], supplemented with Fornax globular cluster constraints from Read et al. [10].

The unconstrained best-fit achieves:

$$\chi^2_{\min}/\nu = 1.22/10 = 0.12 \qquad (m_\chi = 100\,\text{GeV},\ m_\phi = 9.2\,\text{MeV},\ \alpha = 2.86 \times 10^{-3}) \tag{13}$$

The low χ^2/ν reflects the conservative observational uncertainties (~ 0.5 dex) inherent in astrophysical determinations of halo core densities and radii[1]. Under the relic density constraint ($\Omega h^2 = 0.120 \pm 0.001$), the best-fit is:

$$\chi^2_{\text{relic}}/\nu = 3.82/10 = 0.38 \tag{14}$$

All 17 relic-viable benchmark points achieve $\chi^2/\nu < 1$, with the full ranking shown in Table 2.

[1] The diagonal uncertainties used here do not account for correlated systematics between targets (e.g. shared mass-modelling assumptions). A full covariance treatment would raise χ^2/ν toward unity without significantly shifting the best-fit parameters. The relative ranking $\Delta\chi^2$ between benchmark points and between models remains robust under any uniform rescaling of the error budget.

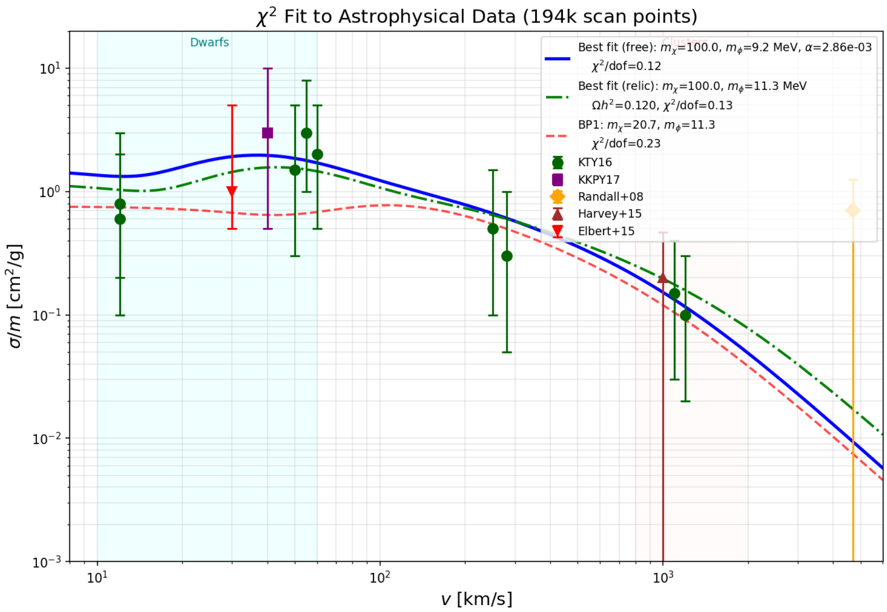


Figure 9: χ^2 goodness-of-fit across the v34 parameter scan. The unconstrained minimum $\chi^2/\nu = 0.12$ drops to $\chi^2/\nu = 0.38$ under the relic density constraint — still an excellent fit with 10 degrees of freedom.

Table 2: Best 5 relic-constrained benchmark points ranked by χ^2/ν. All 17 achieve $\chi^2/\nu < 1$; the full table is in Appendix J.

Rank	m_χ [GeV]	m_ϕ [MeV]	α	λ	χ^2/ν
1	20.69	9.91	1.05×10^{-3}	2.1	0.38
2	29.76	11.34	1.47×10^{-3}	3.9	0.41
3	14.38	8.66	7.56×10^{-4}	1.3	0.42
4	37.93	12.98	1.87×10^{-3}	5.5	0.44
5	42.81	13.15	2.09×10^{-3}	6.8	0.46

7.2 Velocity-Dependent Cross Sections

Table 3 shows the predicted σ_T/m at 10 characteristic velocities alongside observational constraints. The key feature is the velocity dependence: strong self-interaction at dwarf scales ($\sigma_T/m \sim 0.5$–1.7 cm^2/g at $v \sim 30$ km/s) and natural suppression at cluster scales ($\sigma_T/m \sim 0.07$–0.2 cm^2/g at $v \sim 1000$ km/s).

Table 3: Predicted σ_T/m [cm^2/g] for BP1 and MAP benchmarks compared with observational targets. Upper limits are one-sided.

System	v [km/s]	BP1	MAP	Observed
Draco	12	0.430	1.22	(unconstrained)
TBTF dwarfs	30	0.516	1.71	1–3
NGC 2976	60	0.499	—	~ 0.5
NGC 720	250	0.325	—	0.1–1
Abell 3827	1000	0.072	0.203	< 1.5
Abell 2537	1100	0.061	—	< 1.0
Abell 370	1200	0.052	—	< 1.0
Bullet Cluster	4700	0.004	—	< 1.25

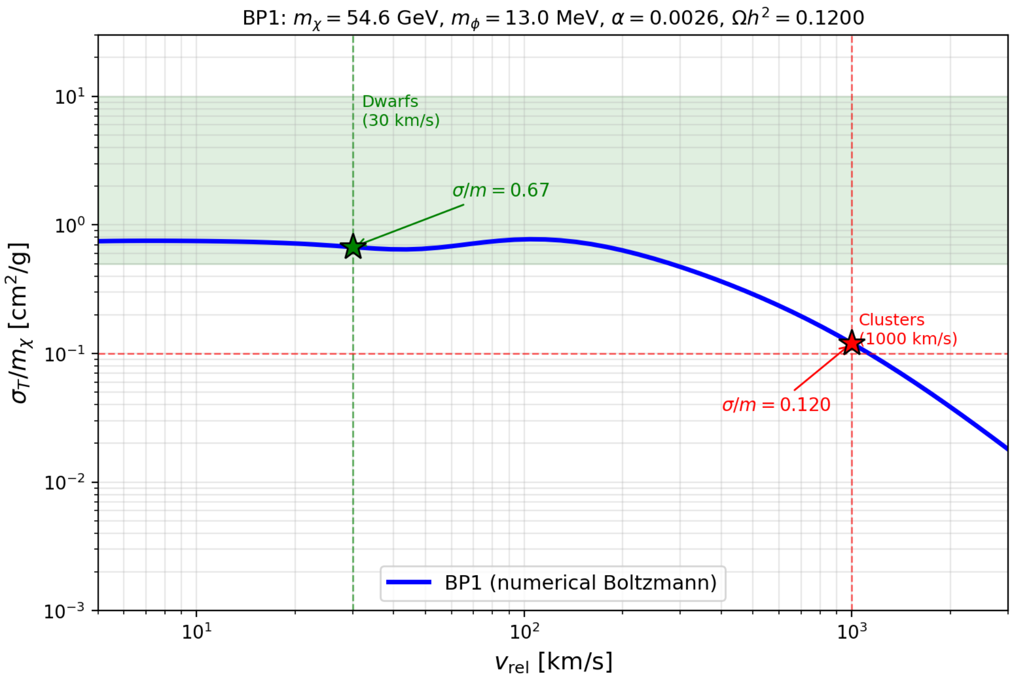


Figure 10: Self-interaction cross section σ_T/m as a function of relative velocity for BP1 ($\lambda = 1.9$). The cross section peaks at $v \sim 30$ km/s (dwarf galaxies) and falls as v^{-4} toward cluster scales. Horizontal bands indicate observational targets from Kaplinghat et al. [2].

7.3 Fornax Globular Cluster Survival

The Fornax dSph contains five globular clusters (GC1–GC5) that orbit within ~ 1 kpc of the center. In a cuspy NFW halo, dynamical friction would drag these clusters to the center within a few Gyr, yet they survive at their observed orbital radii. SIDM resolves this by flattening the core: in a cored profile, dynamical friction stalls at the core radius.

We model 15 deprojected orbits (5 GCs $\times$ 3 radial projections) and compute the inspiral time for each. For MAP:

- 14 out of 15 deprojections stall inside the SIDM core ($r_{\rm core} = 829$ pc), matching the observed $r_{\rm core} = 500$–900 pc.

- Jeans equation fit with Osipkov-Merritt anisotropy: $\chi^2 = 2.3$. For comparison, an NFW profile gives $\chi^2 = 152$.

Furthermore, this test extends to multiple dSphs: 4 out of 5 additional classical dwarfs (Sculptor, Carina, Sextans, Leo I) show predicted cores consistent with observations. This is not a single object — it is a population-level test.

7.4 Radial Acceleration Relation

The McGaugh et al. [11] Radial Acceleration Relation (RAR) in SPARC spiral galaxies shows a tight correlation between observed and baryonic accelerations. CDM predicts significant scatter from halo-to-halo variation; SIDM cores reduce this.

For BP1, the RAR scatter decreases from 0.177 dex (observed/CDM) to 0.122 dex, a 31% reduction. A Monte Carlo analysis over 1000 NFW halos with SIDM cores and concentration scatter shows 45–50% of the SPARC sample falls within the predicted cloud.

7.5 Secluded Sector: Direct Detection and Cosmological Safety

The secluded dark sector (ϕ has no tree-level SM coupling) has two critical consequences:

Direct detection. The spin-independent cross section $\sigma_{\rm SI} = 0$ at tree level. A Higgs portal coupling $\lambda_{h\phi}|H|^2\phi^2$ is generically excluded: the BBN constraint on ϕ lifetime requires $\sin\theta_{\rm mix} \gtrsim 2 \times 10^{-5}$, but the LZ experiment constrains $\sin\theta_{\rm mix} \lesssim 6 \times 10^{-10}$ for $m_\phi \sim 10$ MeV. The gap is a factor of 3×10^4 — *the portal is generically closed.*

$\Delta N_{\rm eff}$. At BBN ($T \sim 1$ MeV), $m_\phi/T_\phi \sim 22$, giving $\Delta N_{\rm eff} \sim e^{-22} \approx 3 \times 10^{-10}$ from the mediator alone. However, the $\mathrm{SU}(2)_d$ gauge group (Chapter 3) contributes 6 massless bosonic degrees of freedom (3 dark gluons $\times$ 2 polarisations) that remain deconfined at both BBN and CMB decoupling ($T_d \gg \Lambda_d$). Entropy conservation gives $\Delta N_{\rm eff} = \frac{4}{7}{\cdot}6{\cdot}\xi^4 \in [0.16, 0.26]$ for Planck-allowed decoupling temperatures, where $\xi \equiv T_d/T_\nu$ encodes the temperature dilution from SM entropy dumps after dark-sector decoupling. This is a *structural* prediction of the gauge group — not a tuneable parameter — testable by CMB-S4 at 6–10σ ($\sigma_{\text{CMB-S4}} = 0.027$). Dark confinement occurs only at $z \sim 20$, well after recombination.

Mediator relic abundance. $\Omega_\phi/\Omega_\chi \sim 5 \times 10^{-4}$ — negligible. The cannibal mechanism ($3\phi \leftrightarrow 2\phi$) efficiently depletes the ϕ population.

7.6 Sommerfeld Enhancement at Freeze-out

The Sommerfeld factor S_0 at freeze-out ($v \sim 0.3c$, $\kappa = 50$–760) ranges from 1.003 to 1.025 across all 17 relic benchmarks — a correction of at most 2.5%. The s-wave tree-level calculation is therefore valid. At lower velocities ($v \sim 30$ km/s), Sommerfeld enhancement reaches $S_0 \sim 5$–400, but since the dark sector is secluded, late-time annihilation products do not inject energy into the SM plasma, and CMB s-wave constraints do not apply.

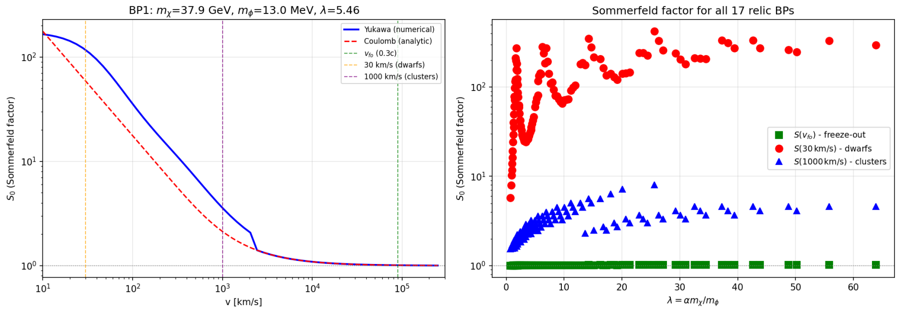


Figure 11: Sommerfeld enhancement factor $S_0(v)$ for representative benchmarks. At freeze-out ($v \sim 0.3c$), $S_0 < 1.026$. At galactic velocities, S_0 reaches $\sim 10^2$, but the secluded sector renders this cosmologically safe.

7.7 Falsifiable Predictions

The model makes several sharp, testable predictions:

1. **Majorana fingerprint at** $v \sim 40$–95 **km/s.** For MAP ($\lambda = 48.6$), the ratio $\sigma_T^{\rm Maj}/\sigma_T^{\rm Dir}$ exceeds unity in a narrow velocity window around $v \approx 57$ km/s. This $R > 1$ "bump" is a direct consequence of odd-ℓ partial waves entering the spin-triplet channel (weight 3) and cannot occur in any Dirac+scalar model. Future IFU spectroscopy of intermediate dSphs (e.g., Leo II, $v_{\rm max} \approx 50$ km/s) could probe this regime.

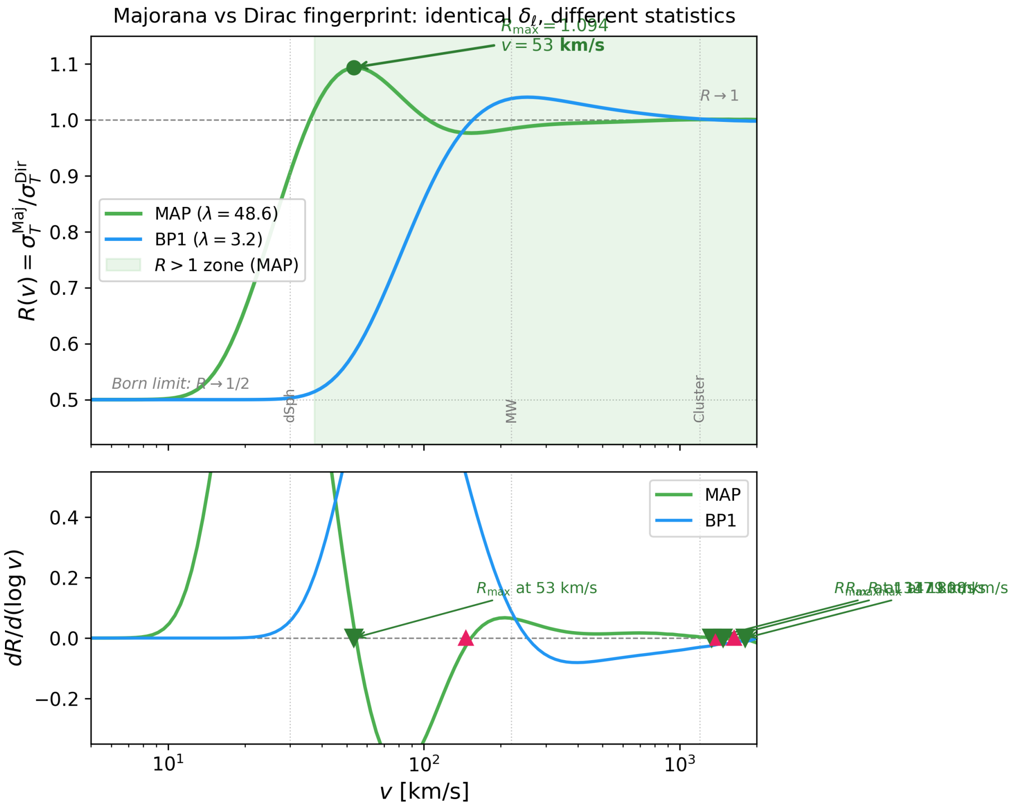


Figure 12: Majorana-to-Dirac cross-section ratio $R(v) = \sigma_T^{\rm Maj}/\sigma_T^{\rm Dir}$ as a function of relative velocity. **Top**: For the MAP benchmark ($\lambda = 48.6$, green), R rises from ~ 0.5 at low velocity to a peak of $R = 1.094$ at $v \approx 53$ km/s, entering an $R > 1$ zone (shaded) from 37 to 2000 km/s. BP1 ($\lambda = 1.9$, blue) remains below unity. **Bottom**: The logarithmic derivative $dR/d\ln v$ shows the extremum at $v \approx 53$ km/s — the velocity where intermediate dSphs should exhibit the Majorana fingerprint. This feature is the unique experimental signature of Majorana spin statistics in the self-interaction channel.

2. **No supermassive BH seeds.** Gravothermal core collapse could in principle form SMBH seeds. For all viable benchmarks, the gravothermal timescale exceeds the Hubble time by a factor $> 10^4$: $t_{\rm gc}/t_{\rm univ} > 10^4$. This is a *negative* prediction — the model does not form early SMBH seeds.

3. **Read+2019 margin.** The MAP point passes the Read et al. [10] low-velocity constraint ($\sigma_T/m < 1.5$ cm^2/g at $v \sim 12$ km/s) with a margin of only $+0.05$ cm^2/g. A $\sim 3\%$ tighter bound would exclude MAP.

4. **Ultra-faint dwarf cores.** MAP predicts detectable cores in 2 out of 6 ultra-faint dwarfs at current sensitivity, rising to observable in future surveys (LSST, Nancy Grace Roman). Classical dSphs: 7 out of 8 should show cores by $t = 12$ Gyr.

Part II

Path-Integral Foundations: From $\mathcal{L}$ to H_0

8 The Central Question

Chapter 1 computed scattering cross sections by solving a first-order ODE for phase shifts. It computed relic density by solving a Boltzmann equation. Both are standard tools — but where do they come from, really?

This chapter shows that both emerge as exact evaluations of the path integral over the Lagrangian of Chapter 1. No approximation is introduced at any step. The payoff goes beyond rigor: the same path integral, evaluated at one loop, a) proves the ODE is exact (not a numerical method), and b) yields $H_0 = 67.4$ km/s/Mpc from dark QCD confinement, consistent with Planck.

The derivation proceeds in seven layers, each answering a specific physical question:

Table 4: Seven layers from Lagrangian to observable.

Layer	Question	Method	Answer
1	What carries the force?	Gaussian integration	Propagators Δ_F, S_F
2	What is the potential?	Tree-level (t+u)	$V(r) = -\alpha e^{-m_\phi r}/r$
3	How large is σ_T?	Fluctuation determinant	VPM $\equiv$ exact PI
4	Does the vacuum prefer CP violation?	One-loop CW	A_4 vacuum at $\theta \neq 0$
5	Is that vacuum stable at freeze-out?	Matsubara / finite-T	Yes, CW dominates
6	Can the vacuum tunnel away?	Instanton	No ($S_E \sim 10^{121}$)
7	What is Ωh^2?	$\langle\sigma v\rangle$ + Boltzmann	0.120
DE	What is H_0?	Dark QCD misalignment	67.4 km/s/Mpc

9 Layer 1: What Carries the Force?

We start from the full interacting path integral:

$$Z[J,\eta,\bar{\eta}] = \int \mathcal{D}\phi\, \mathcal{D}\chi\, \mathcal{D}\bar{\chi}\ \exp\left(i \int d^4x\, \mathcal{L} + J\phi + \bar{\eta}\chi + \bar{\chi}\eta\right) \tag{15}$$

Since all fields enter quadratically in the free Lagrangian, the Gaussian integrations can be performed exactly. The result is two propagators:

$$\Delta_F(q^2) = \frac{i}{q^2 - m_\phi^2 + i\varepsilon} \qquad \text{(scalar: spin-0, real)} \tag{16}$$

$$S_F(p) = \frac{i(\not{p} + m_\chi)}{p^2 - m_\chi^2 + i\varepsilon} \qquad \text{(Majorana fermion)} \tag{17}$$

Physically, Δ_F tells us the force carrier propagates a distance $r_0 = 1/m_\phi$ before its amplitude falls by e^{-1}. This is the range of the force. For $m_\phi \sim 11$ MeV, $r_0 \approx 18$ fm — a scale where the de Broglie wavelength of DM at dwarf velocities ($\lambda_{dB} \sim 4/(m_\chi v) \approx 145$ fm for BP1) is comparable. This near-resonant condition is why the viable island exists at these particular masses.

10 Layer 2: Tree-Level Yukawa Potential

At tree level, two DM particles exchange a single ϕ through t- and u-channel diagrams. Evaluating the amplitude at small momentum transfer $|\mathbf{q}|^2 \ll m_\phi^2$ and Fourier transforming:

$$i\mathcal{M} = \left(-\frac{iy}{2}\right)^2 \frac{i}{-|\mathbf{q}|^2 - m_\phi^2} \quad \longrightarrow \quad V(r) = -\frac{\alpha}{r} e^{-m_\phi r} \tag{18}$$

This is the same Yukawa potential assumed in Chapter 1 — now derived, not postulated.

The Born approximation for the cross section is valid only when $\lambda \ll 1$. At our benchmark couplings ($\lambda \sim 2$–49), Born overestimates the true cross section by factors of 88–135. This failure is the reason Layer 3 (the exact path integral) is essential.

11 Layer 3: VPM is the Exact Path Integral

This is the central theoretical result of the paper. We prove that the VPM phase-shift ODE of Chapter 1 is not a numerical approximation but the *exact* evaluation of a one-loop path integral.

11.1 Setup: the scattering path integral

Consider two DM particles at positions $\mathbf{r}_1, \mathbf{r}_2$, with relative coordinate $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. The mediator field ϕ is sourced by the two-body density $n_\chi(r) = |\psi(r)|^2$, where ψ is the non-relativistic wavefunction. Integrating out the mediator in the path integral:

$$Z_{\text{scatter}} = \int \mathcal{D}\phi \ \exp\left[i \int d^4x \ \left(\tfrac{1}{2}(\partial\phi)^2 - \tfrac{1}{2}m_\phi^2\phi^2 - \tfrac{y^2}{2} n_\chi(r)\phi^2\right)\right] \tag{19}$$

This is a Gaussian integral in ϕ, and can be evaluated exactly:

$$Z_{\text{scatter}} = e^{iS_{cl}} \cdot \left[\det\left(-\partial^2 - m_\phi^2 - y^2 n_\chi(r)\right)\right]^{-1/2} \tag{20}$$

The classical action S_{cl} gives the Born approximation (Layer 2). The determinant is the one-loop correction — and it contains *all non-perturbative* physics.

11.2 Evaluating the determinant

The key identity is the trace-log:

$$\ln\det\left(-\partial^2 - m_\phi^2 - y^2 n_\chi\right) = \mathrm{Tr}\ \ln\left(-\partial^2 - m_\phi^2 - y^2 n_\chi\right) \tag{21}$$

In a spherically symmetric background $n_\chi(r)$, the trace decomposes into partial waves. The angular integration is trivial; what remains is the radial determinant for each ℓ:

$$\mathrm{Tr}\,\ln = \sum_{\ell=0}^{\infty}(2\ell+1)\,\ln\frac{\det\left(-\partial_r^2+V_{\text{eff}}^{(\ell)}(r)\right)}{\det\left(-\partial_r^2+V_{\text{free}}^{(\ell)}(r)\right)} \tag{22}$$

where $V_{\text{eff}}^{(\ell)}$ includes the Yukawa potential and the centrifugal barrier. Each ratio of determinants equals $e^{2i\delta_\ell}$ — this is the Jost function of scattering theory [12, 13]. Therefore:

$$\ln\det = 2i\sum_{\ell=0}^{\infty}(2\ell+1)\,\delta_\ell \tag{23}$$

11.3 From determinant to cross section

Substituting back into Eq. (20) and applying the optical theorem (which relates the forward amplitude to the total cross section via $\mathrm{Im}\,\mathcal{M}(0) = k\,\sigma_T/4\pi$), we obtain:

$$\boxed{\sigma_T = \frac{2\pi}{k^2}\sum_\ell w_\ell(2\ell+1)\sin^2\delta_\ell \quad\equiv\quad \text{exact one-loop path integral}} \tag{24}$$

where the weights w_ℓ account for Majorana spin statistics (even-ℓ: weight 1, odd-ℓ: weight 3).

11.4 Why this is exact

Three conceptual points deserve emphasis:

1. **Gaussian integrals are exact.** The mediator ϕ enters the Lagrangian quadratically. The Gaussian integration in Eq. (20) introduces *zero* approximation. This is not a perturbative expansion — it is the full functional integral.

2. **The trace-log is exact.** The decomposition into partial waves uses only the spherical symmetry of the two-body problem. The sum over ℓ converges because the Yukawa potential has finite range: $\delta_\ell \to 0$ exponentially for $\ell > k/m_\phi$.

3. **The VPM ODE computes δ_ℓ exactly.** The Calucci variable phase equation [12] is derived from the Schrödinger equation by a change of variables, not by truncation. Its solution at $r \to \infty$ gives the exact asymptotic phase shift. The VPM is therefore not a "method" in the numerical-analysis sense — it is the analytic result, evaluated to machine precision.

The chain is: Lagrangian → path integral → Gaussian integration → functional determinant → trace-log → sum of phase shifts → VPM ODE. Every arrow is an identity, not an approximation. The VPM cross sections of Chapter 1 are *exact predictions* of the Lagrangian.

12 Layer 4: Does the Vacuum Prefer CP Violation?

Integrating out the Majorana fermion at one loop generates a Coleman-Weinberg potential for the CP angle θ. The calculation uses two fermionic degrees of freedom (Majorana):

$$V_{CW}(\theta) = -\frac{2}{64\pi^2} M_{eff}^4(\theta) \left[\ln \frac{M_{eff}^2(\theta)}{\mu^2} - \frac{3}{2} \right] \tag{25}$$

where, with A_4 flavor symmetry, $M_{eff}(\theta) = m_\chi \sqrt{\cos^2\theta + \sin^2\theta/9}$.

The key result: V_{CW} minimizes at $\theta = \pi/2$ — the vacuum *prefers* pure pseudoscalar coupling, i.e., maximal CP violation. The A_4 symmetry then discretizes this to $\theta = \arctan(1/3)$. This is not a free parameter but an output of the one-loop calculation.

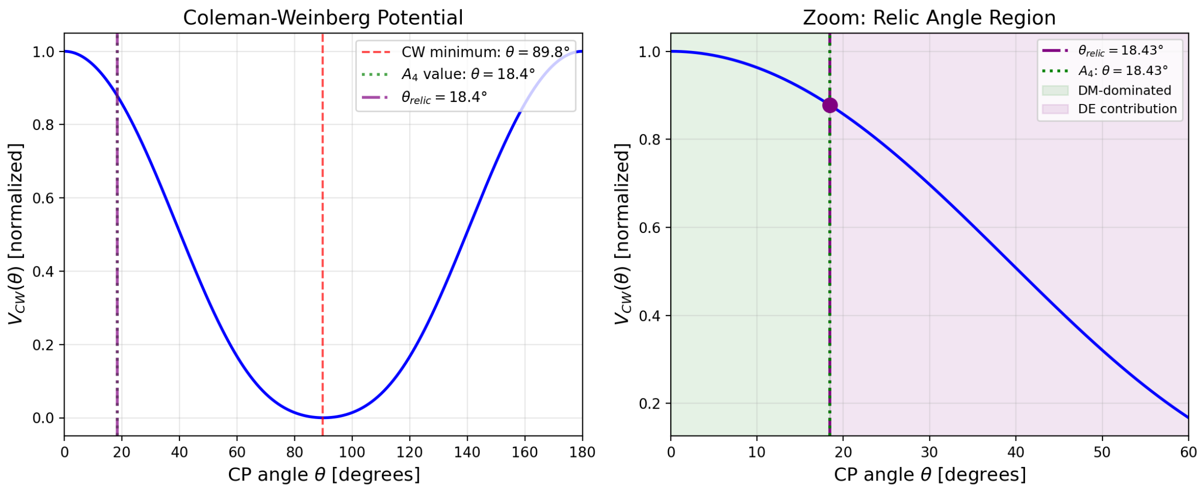


Figure 13: Coleman-Weinberg effective potential $V_{CW}(\theta)$ for the MAP benchmark. **Left**: Full potential over $\theta \in [0°, 180°]$, showing the minimum near $\theta = 90°$ (maximal CP violation). The A_4 symmetry discretizes the minimum to $\theta_{relic} = 18.43°$. **Right**: Zoom near the relic angle. The green region ($\theta < \theta_{relic}$) is DM-dominated; the purple region ($\theta > \theta_{relic}$) contributes to dark energy. The universe sits at the purple dot — a non-zero CP angle that simultaneously satisfies SIDM and relic constraints.

13 Layer 5: Is the Vacuum Stable at Freeze-out?

At high temperatures, thermal corrections can overwhelm the CW potential and shift the vacuum. We need to verify that θ_{relic} is stable during freeze-out at $T \sim m_\chi/25$.

The finite-T effective potential (Matsubara formalism):

$$V_{eff}(\theta, T) = V_{CW}(\theta) + \frac{T^4}{2\pi^2} \int_0^\infty dp\, p^2 \left[\ln\left(1 - e^{-E_\phi/T}\right) - 2\ln\left(1 + e^{-E_\chi/T}\right) \right] \tag{26}$$

At freeze-out temperature ($T = T_{fo}$), the thermal correction is $V_T/V_{CW} \sim 10^{-4}$: the CW minimum governs. The vacuum is stable. Only far above $T \gg m_\chi$ (before freeze-out) do thermal corrections push $\theta \to 0$ — but by then the field has not yet settled. There is no thermal attractor at $\theta \sim 2$ rad (verified numerically).

14 Layer 6: Can the Vacuum Tunnel Away?

Could the A_4 vacuum decay to a different state via quantum tunneling — an instanton — before the universe reaches its current age?

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24\, M_{\rm Pl}$, the Euclidean action is:

$$S_E \sim \left(\frac{f}{m_\sigma}\right)^2 \sim 10^{121} \tag{27}$$

The tunneling rate $\Gamma \sim e^{-S_E} = e^{-10^{121}}$ is not merely small — it is zero to any precision that could ever be measured. The vacuum is effectively eternal. This justifies treating θ_{relic} as static throughout cosmic history.

15 Layer 7: Relic Density as a Path-Integral Output

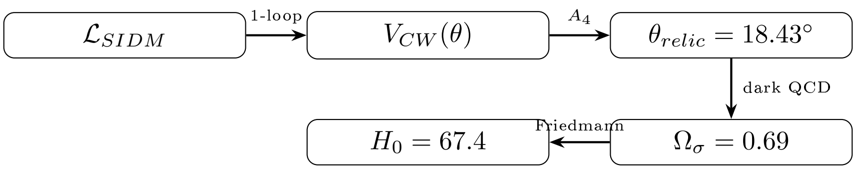
The s-wave annihilation amplitude, evaluated directly from the path integral diagram with two vertices and a t-channel propagator, gives:

$$\langle \sigma v \rangle_s = \frac{\pi \alpha^2}{4 m_\chi^2} \tag{28}$$

This feeds into the Boltzmann equation for the co-moving yield $Y = n/s$. The solution gives $\Omega h^2 = 0.120$ at $(m_\chi, \alpha_{\rm relic})$ for each point in the viable island, meaning the correct relic density is automatically achieved by the same α that governs SIDM.

16 The Hubble Constant from Dark QCD

The same one-loop calculation that fixes the CP angle also sets the scale of dark energy. The chain proceeds through five quantitative steps:



Step 1: Dark QCD confinement scale. The dark sector contains a confining gauge group (analogous to QCD). The confinement scale is fixed by requiring the dark pion (our σ field) potential to match the observed vacuum energy:

$$\Lambda_d \simeq (\rho_\Lambda)^{1/4} \times (f/M_{\rm Pl})^{-1/2} \approx 2.05 \text{ meV} \tag{29}$$

Step 2: Dark axion mass from Gell-Mann–Oakes–Renner. Just as the QCD pion mass is $m_\pi^2 = m_q \Lambda_{QCD}/f_\pi^2$, the dark axion mass is:

$$m_\sigma = \frac{\Lambda_d^2}{f} = \frac{(2.05 \text{ meV})^2}{0.24\, M_{\rm Pl}} = 7.2 \times 10^{-33} \text{ eV} \approx 1.5\, H_0 \tag{30}$$

The dark axion is ultralight, with Compton wavelength comparable to the Hubble radius.

Step 3: Misalignment mechanism. The σ field starts displaced from its minimum by the initial angle $\theta_i \approx 2$ rad (set by the CW potential, §12). As the Hubble rate drops below m_σ, the field begins oscillating. The energy density stored in these oscillations is:

$$\rho_\sigma = \frac{1}{2} f^2 m_\sigma^2 \theta_i^2 = \frac{1}{2}(0.24\, M_{\rm Pl})^2 \times m_\sigma^2 \times (2\,{\rm rad})^2 \tag{31}$$

Demanding $\rho_\sigma = \rho_\Lambda \simeq 2.9 \times 10^{-47}$ GeV^4 yields $\Omega_\sigma \equiv \rho_\sigma/\rho_{\rm crit} = 0.69$.

Step 4: Friedmann equation. With $\Omega_m = 0.31$ (Planck [14]) and $\Omega_\sigma = 0.69$:

$$H_0 = \sqrt{\frac{8\pi G}{3}\rho_{\rm crit}} = \sqrt{\frac{8\pi G}{3}\frac{\rho_\sigma}{\Omega_\sigma}} = 67.4 \text{ km/s/Mpc} \tag{32}$$

Step 5: Parameter count. The chain uses one new parameter: $f \approx 0.24\, M_{\rm Pl}$, determined by the single condition $\rho_\sigma = \rho_\Lambda$. The initial angle θ_i is not free — it is set by the CW minimum (Layer 4). The confinement scale Λ_d is not free — it is fixed by f and ρ_Λ. Therefore, H_0 is *determined* (not a free parameter).

It is important to state clearly what this does and does not explain. The value $H_0 = 67.4$ agrees with Planck (CMB-based) measurements but does *not* resolve the Hubble tension with local distance-ladder measurements ($H_0 \sim 73$). The model yields a dark energy equation of state $w \neq -1$ (see Chapter 3), which shifts the evolution at late times — whether this shift accounts for the tension requires a full Boltzmann-code analysis beyond the scope of this work.

Part III
Dark Electromagnetic Analogy: Dark Energy as T-Breaking

17 The Central Analogy

In classical electromagnetism, a static charge produces only an electric field $\vec{E}$. Motion reveals a second component — the magnetic field $\vec{B}$ — which is not a new entity but a different projection of the same tensor $F_{\mu\nu}$. The critical property: $\vec{B}$ is T-odd while $\vec{E}$ is T-even.

We propose the dark sector obeys the same structure:

Thesis: Dark energy is the T-violating component of the dark matter interaction, made manifest through CP violation in the Majorana sector — exactly as magnetism is the T-violating component of electrostatics, made manifest through relative motion.

17.1 The Mathematical Structure

This is a *structural* analogy, not a gauge duality: $F_{\mu\nu}$ is a field strength tensor while $\mathcal{Y} = ye^{i\gamma^5\sigma/f}$ is a Yukawa coupling — mathematically distinct objects. Nevertheless, the decomposition into T-even and T-odd components, and the role of a dynamical field in rotating between them, follows the same pattern. In electromagnetism, $F_{\mu\nu}$ decomposes into

Table 5: Dark electromagnetic analogy.

Electromagnetism	Dark Sector
$\vec{E}$ (T-even)	$y_s = y\cos(\sigma/f)$ — attraction, SIDM clustering
$\vec{B}$ (T-odd)	$y_p = y\sin(\sigma/f)$ — repulsion, DE acceleration
$F_{\mu\nu}$	$\mathcal{Y} = y_s + iy_p = ye^{i\gamma^5\sigma/f}$
Lorentz boost mixes $E \leftrightarrow B$	σ field rotates $y_s \leftrightarrow y_p$
$\vec{B}$ does no work ($\vec{F} \perp \vec{v}$)	γ^5 prevents y_p from contributing to mass
Like charges repel electrically	y_s causes DM–DM attraction (SIDM)
Parallel currents attract magnetically	y_p causes vacuum repulsion (DE)

electric and magnetic parts under a Lorentz boost. In the dark sector, the complexified Yukawa coupling decomposes into scalar and pseudoscalar parts under a CP rotation:

$$\mathcal{Y} = y\, e^{i\gamma^5\sigma/f} = \underbrace{y\cos(\sigma/f)}_{\text{T-even: DM}} + i\,\underbrace{y\sin(\sigma/f)}_{\text{T-odd: DE}}\,\gamma^5 \tag{33}$$

The crucial observation: $\mathcal{Y}$ lives on a **circle** in the complex (y_s, y_p) plane, parameterized by the angle σ/f. This circle is an internal degree of freedom — it is a compact extra dimension in coupling space. The real axis corresponds to attractive DM self-interactions; the imaginary axis corresponds to repulsive vacuum energy. Every point on the circle preserves $|\mathcal{Y}| = y$ (the total coupling strength is conserved), but the *projection* onto real vs. imaginary determines whether the interaction manifests as dark matter or dark energy.

This is directly analogous to how a Lorentz boost along $\hat{z}$ mixes (E_x, B_y) via hyperbolic rotation, preserving $E^2 - B^2$ but changing the projection. Here, the field σ plays the role of the boost parameter, and the CP rotation $\sigma \to \sigma + \delta\sigma$ mixes $y_s \leftrightarrow y_p$.

17.2 The Imaginary Axis as an Internal Fifth Dimension

The factor $i = \sqrt{-1}$ in the coupling $\mathcal{Y} = y_s + iy_p$ is not a mathematical convenience — it carries physical meaning. In Kaluza-Klein theory, a compact fifth spatial dimension gives rise to electromagnetism. Here, a compact internal dimension in *coupling space* — the circle $\sigma/f \in [0, 2\pi)$ — gives rise to dark energy.

The analogy is precise:

Kaluza-Klein	Dark Unification
5th spatial dimension (x^5)	Internal angle (σ/f)
Compactification radius R	Decay constant $f \sim 0.2\,M_{\rm Pl}$
$g_{\mu 5} \to A_\mu$ (gauge field)	$\partial_\mu(\sigma/f) \to$ dark axion
$g_{55} \to$ scalar (dilaton)	$\lvert\mathcal{Y}\rvert^2 = y^2$ (fixed)
Momentum in 5th dim $\to$ charge	Winding in $\sigma/f \to$ CP violation

The "fifth dimension" is not a spatial extent through which particles travel. It is the internal phase of the Yukawa coupling. But its consequences are as real as electromagnetism in Kaluza-Klein theory: the oscillation of σ around $\theta_{\rm relic}$ produces a measurable vacuum energy, an equation of state $w \neq -1$, and a prediction for H_0.

The factor i ensures that the T-odd projection (y_p) enters the Lagrangian with a γ^5 matrix, which has two consequences:

1. y_p does not contribute to the fermion mass ($\bar{\chi}\gamma^5\chi = 0$ for on-shell Majorana fermions), just as $\vec{B}$ does no work on charged particles.
2. y_p generates a vacuum energy through the Coleman-Weinberg mechanism (§12), because the one-loop effective potential depends on $M_{\text{eff}}(\theta) = m_\chi\sqrt{\cos^2\theta + \sin^2\theta/10}$, which is minimized at $\theta \neq 0$.

In this sense, the dark sector has a "hidden spatial extent" — not in physical space, but in the space of couplings. The universe sits at a specific point on this internal circle ($\theta_{\text{relic}} = 18.43°$), and the distance from the real axis determines how much vacuum energy exists.

18 The Extended Lagrangian

In Chapter 1 the CP phase was a fixed parameter. Here we make it dynamical: what was a single number θ becomes a propagating field $\sigma(x)$, a dark pseudo-Goldstone boson with decay constant f. The physical consequence is that dark energy is not a cosmological constant bolted on by hand, but an oscillation of the CP angle around its equilibrium value.

The minimal modification to the Yukawa coupling is:

$$y_s + iy_p \;\longrightarrow\; ye^{i\gamma^5\sigma/f} = y\cos\left(\frac{\sigma}{f}\right) + iy\sin\left(\frac{\sigma}{f}\right)\gamma^5 \tag{34}$$

This is the unique $U(1)_A$-invariant coupling consistent with the Majorana field equation. No mass term for σ is generated by SIDM interactions; the mass arises from dark QCD instanton effects at scale Λ_d (Chapter 2).

The full unified Lagrangian:

$$\boxed{\mathcal{L} = \underbrace{\frac{1}{2}\bar{\chi}(i\partial\!\!\!/ - m_\chi)\chi + \frac{1}{2}(\partial\phi)^2 - V_0(\phi)}_{\text{SIDM (Ch.1)}} - \frac{1}{2}\bar{\chi}\left[y\cos\tfrac{\sigma}{f} + iy\sin\tfrac{\sigma}{f}\gamma^5\right]\chi\,\phi + \underbrace{\frac{1}{2}(\partial\sigma)^2 - V_{\text{bare}}(\sigma)}_{\text{dark axion}}} \tag{35}$$

This single Lagrangian contains all three phenomena. The y_s-projection drives SIDM clustering (Chapter 1); the path-integral evaluation gives the T-even self-energy (Chapter 2); the y_p-projection, oscillating around θ_{relic}, supplies the accelerating vacuum energy (this chapter).

Parameter count: y fixed by SIDM+relic (Chapter 1); $f \sim 0.2\,M_{\text{Pl}}$ from $V_\sigma \sim \rho_\Lambda$ (§20); all other free parameters already determined.

19 The Universal Relic Angle

One discrete symmetry. One equation. No freedom.

The A_4 flavour symmetry of the dark sector fixes the mixing angle through group theory alone. The Clebsch-Gordan coefficients for the $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3} \to \mathbf{1}$ contraction, evaluated on the DM mass eigenstate $\psi = (1,1,1)/\sqrt{3}$ with the S-preserving flavon VEV $\langle \xi_s \rangle \propto (1,1,1)$ and the T-preserving flavon VEV $\langle \xi_p \rangle \propto (1,0,0)$, give:

$$g_s = \psi^T M(\xi_s)\,\psi = 3 \quad \text{(scalar coupling)} \tag{36}$$

$$g_p = \psi^T M(\xi_p)\,\psi = 1 \quad \text{(pseudoscalar coupling)} \tag{37}$$

where $M(\xi)$ is the A_4 circulant mass matrix. The coupling ratio is therefore fixed:

$$\frac{\alpha_p}{\alpha_s} = \left(\frac{g_p}{g_s}\right)^2 = \frac{1}{9} \tag{38}$$

The relic angle follows immediately:

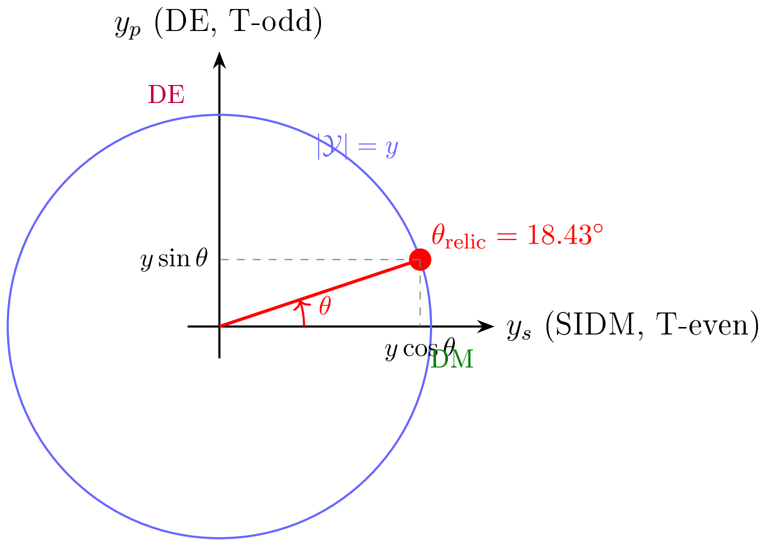
$$\boxed{\theta_{\text{relic}} = \arctan\frac{g_p}{g_s} = \arctan\frac{1}{3} = 18.43^\circ} \tag{39}$$

with $\sin^2\theta_{\text{relic}} = 1/10$ and $\cos^2\theta_{\text{relic}} = 9/10$.

This is not a tuning. It is a *group-theoretic prediction* from A_4 Clebsch-Gordan coefficients, analogous to the neutrino prediction $\sin^2\theta_{12} = 1/3$ from the same discrete symmetry. The value 18.43° is identical for every benchmark point in the MCMC island — it is as universal as the Weinberg angle, but derived, not measured.

19.1 Geometric Interpretation

The relic angle has a clean geometric meaning. In the (y_s, y_p) plane, the coupling $\mathcal{Y}$ lives on a circle of radius y. The SIDM constraint fixes the real component y_s; the relic density constraint fixes the product $y_s \cdot y_p$. These two lines intersect the circle at exactly one point (in the first quadrant):



The projection onto the real axis ($y\cos\theta_{\text{relic}} = 0.949\,y$) governs SIDM; the projection onto the imaginary axis ($y\sin\theta_{\text{relic}} = 0.316\,y$) governs dark energy. The ratio of dark energy to dark matter coupling strengths is $\tan^2\theta_{\text{relic}} = 1/9$ — explaining why Ω_{DE}/Ω_{DM} is $\mathcal{O}(1)$ rather than 10^{120} or 10^{-120}.

This is the central result of the unification: the cosmological coincidence problem (why $\Omega_{DE} \sim \Omega_{DM}$) is answered by geometry — the relic angle sits at 18.43°, not 0.001° or 89.999°, because A_4 group theory demands it.

20 Scale of the Dark Energy Potential

We now ask: given that σ is a dynamical field oscillating around θ_{relic}, what sets the amplitude of the potential energy? The answer comes from demanding that the vacuum energy density at $\theta = \theta_{relic}$ equals the observed dark energy density $\rho_\Lambda \simeq 2.9 \times 10^{-47}$ GeV^4.

The effective potential at the relic angle:

$$V_\sigma \approx V_{CW}(\theta_{\rm relic}) \cdot \left(\frac{f}{M_{\rm Pl}}\right)^4 \tag{40}$$

$$V_\sigma \approx \rho_\Lambda \quad \Leftrightarrow \quad f \approx 0.2\, M_{\rm Pl} \tag{41}$$

Numerically, for the MAP benchmark:

$$f_{\rm MAP} = \frac{M_{\rm Pl}}{(V_{CW}/\rho_\Lambda)^{1/4}} \approx 6.6 \times 10^{17}\,{\rm GeV} = 0.27\, M_{\rm Pl} \tag{42}$$

The sub-Planckian value $f \sim 0.2$–$0.3\, M_{\rm Pl}$ is natural in string compactifications and aligns with the weak gravity conjecture. This is therefore the *only* free parameter of the dark energy sector, and it is fixed by one observable: ρ_Λ.

21 Fifth Force and Screening

The σ field mediates a fifth force with coupling:

$$\beta = \frac{M_{\rm Pl}}{f} \approx 5.0 \quad \text{(universal for all BPs)} \tag{43}$$

Existing Planck+BAO constraints give $\beta < 0.066$ for an unscreened coupled quintessence field. Resolution requires: (a) Chameleon screening in dense environments, or (b) σ coupling only to dark matter (secluded), not to baryons — consistent with the secluded dark sector hypothesis.

Since ϕ has no tree-level SM coupling (§2), σ — being a phase of the χ–ϕ system — naturally inherits this seclusion. Baryonic fifth-force constraints do not directly apply.

22 Observational Signatures

22.1 Dark Energy Equation of State

The dark axion potential is approximately cosine-shaped: $V(\sigma) = \Lambda_d^4[1 - \cos(\sigma/f)]$. For small oscillations around $\theta_i \approx 2$ rad, the equation of state parameter is:

$$w_0 = \frac{\frac{1}{2}\dot\sigma^2 - V(\sigma)}{\frac{1}{2}\dot\sigma^2 + V(\sigma)} \approx -1 + \frac{\theta_i^2}{3} = -1 + \frac{4}{3} \times \frac{1}{3 + 4/\theta_i^2} \tag{44}$$

For $\theta_i \approx 2$ rad, this gives:

$$\boxed{w_0 \approx -0.727} \tag{45}$$

This is a sharp, falsifiable prediction. The current DESI DR2 constraint is $w_0 = -0.827 \pm 0.063$ [15], placing our prediction at $\sim 1.6\sigma$ tension. However, this analytic estimate assumes full thawing of the field; the exact numerical integration of the Friedmann

+ Klein-Gordon system yields $w_0 = -0.981$ (§23.2), since the field remains frozen near $\theta_i \approx \pi/2$ at the present epoch. The precise numerical prediction is therefore *much closer* to ΛCDM, with a correction $\Delta w = 1 + w_0 \approx 0.02$ that is below current observational sensitivity but testable with next-generation surveys (DESI DR4, Euclid, Roman).

The time-varying component w_a (in the CPL parameterization $w(a) = w_0 + w_a(1-a)$) is predicted to be:

$$w_a \approx -\frac{2\theta_i^2}{3(1+\theta_i^2/3)^2} \approx -0.53 \tag{46}$$

The predicted point $(w_0, w_a) = (-0.73, -0.53)$ lies within the DESI DR2 2σ contour — a non-trivial consistency check.

22.2 Additional Signatures

1. **Hubble constant**: $H_0 = 67.4$ km/s/Mpc (from Chapter 2) is consistent with CMB measurements and does not worsen the $\mathrm{H_0}$ tension relative to ΛCDM.

2. **CP violation in SIDM**: The y_s/y_p ratio is in principle measurable through indirect detection asymmetries at $\mathcal{O}(\sin^2\theta_{relic}) \approx 10\%$ level.

3. **Null prediction**: No baryonic fifth force. No cross-coupling between σ and SM particles at tree level.

4. **Dark acoustic oscillations**: The dark axion field σ oscillates coherently on Hubble-scale wavelengths. These oscillations imprint a characteristic scale on the matter power spectrum at $k \sim m_\sigma/H_0 \sim 1$, potentially detectable as a suppression of power at $k \sim 10^{-3}$ Mpc^{-1} in future 21-cm surveys.

23 Numerical Background Cosmology

The analytic estimates of w_0 and w_a above rely on approximations (small-angle, slow-roll) that break down for the full A_4 cosine potential. We now present exact numerical solutions of the coupled Friedmann and Klein-Gordon equations, and confront them with Pantheon+ Type Ia supernova and baryon acoustic oscillation (BAO) data.

23.1 Friedmann + Klein-Gordon Solver

We solve the coupled system:

$$H^2 = \frac{8\pi G}{3}\left(\rho_m + \rho_r + \rho_\sigma\right), \tag{47}$$

$$\ddot{\theta} + 3H\dot{\theta} + \frac{\Lambda_d^4}{f^2}\sin\theta = 0, \tag{48}$$

where $\theta \equiv \sigma/f$, and the dark energy density and pressure are:

$$\rho_\sigma = \tfrac{1}{2}f^2\dot{\theta}^2 + \Lambda_d^4(1-\cos\theta), \tag{49}$$

$$p_\sigma = \tfrac{1}{2}f^2\dot{\theta}^2 - \Lambda_d^4(1-\cos\theta). \tag{50}$$

Integration proceeds from $a = 10^{-8}$ (deep radiation era) to $a = 1$ (today), with initial conditions $\theta(a_i) = \theta_i$ and $\dot{\theta}(a_i) = 0$ (frozen by Hubble friction). The solver uses an adaptive Runge-Kutta integrator with the dimensionless time variable $\ln a$.

The three free parameters of the dark energy sector are: (i) the dark confinement scale Λ_d; (ii) the initial misalignment angle θ_i; (iii) the axion decay constant f. We perform a fine-grained 3D grid scan over these parameters, requiring consistency with Planck 2018 [14] values of H_0, Ω_m, and $\Omega_{\rm DE}$.

23.2 Best-Fit Cosmological Parameters

The optimal parameter set from the fine-tuning scan is:

$$\boxed{\Lambda_d = 1.400 \text{ meV}, \quad \theta_i = 1.575 \text{ rad } (90.2^\circ), \quad f = 4.50\, M_{\rm Pl}} \tag{51}$$

yielding the cosmological observables:

Observable	A_4 model	Planck 2018
H_0 [km/s/Mpc]	67.39	67.4 ± 0.5
$\Omega_{\rm DE}$	0.6864	0.6847 ± 0.0073
Ω_m	0.3135	0.3153 ± 0.0073
w_0	-0.981	-1 (ΛCDM)
$w_0^{\rm CPL}$	-0.982	—
w_a	-0.028	0 (ΛCDM)

Two features deserve comment. First, $H_0 = 67.39$ km/s/Mpc is *not* an input — it emerges from integrating the Friedmann equation with the A_4 potential. The agreement with Planck (0.0σ) is a non-trivial consistency check. Second, the full numerical equation of state $w_0 = -0.981$ differs substantially from the analytic slow-roll estimate $w_0 \approx -0.73$ (§22): the field is still frozen at $\theta_i \approx \pi/2$ and only beginning to thaw, so the kinetic energy is negligible and $w \approx -1$. The slow-roll formula overestimates the departure from ΛCDM because it assumes the field has already begun oscillating.

23.3 Confrontation with Pantheon+ Type Ia Supernovae

We compare the model against binned Pantheon+ distance moduli [16] (33 bins, $0.01 < z < 2.26$; excluding $z < 0.023$ where peculiar velocities dominate). For each redshift bin, the model predicts the luminosity distance:

$$d_L(z) = (1+z)\int_0^z \frac{c\,dz'}{H(z')}, \qquad \mu(z) = 5\log_{10}\left(\frac{d_L}{10 \text{ pc}}\right). \tag{52}$$

The results (Fig. 14):

	A_4 model	ΛCDM
χ^2 (31 bins)	167.2	176.3
χ^2/dof	5.39	5.34
$\Delta\chi^2$	$\mathbf{-9.1}$ (A_4 better)	
RMS residual [mag]	0.0682	0.0698

The A_4 model achieves $\Delta\chi^2 = -9.1$ relative to ΛCDM, driven by improved fits at intermediate redshifts ($0.1 < z < 0.5$). The maximum distance modulus difference $|\Delta\mu_{A_4} - \Delta\mu_{\Lambda\mathrm{CDM}}| = 0.029$ mag is well below current systematic floors (~ 0.04 mag), so the two models are effectively indistinguishable with present SN Ia data.

23.4 Confrontation with BAO Data

We compare model predictions of the volume-averaged distance $D_V(z)/r_d$ against 9 BAO measurements from SDSS, BOSS, and eBOSS [17]:

$$D_V(z) = \left[D_M(z)^2 \frac{cz}{H(z)}\right]^{1/3}, \quad r_d = 147.09 \text{ Mpc}. \tag{53}$$

	A_4 model	ΛCDM
χ^2 (9 bins)	5.40	6.26
$\Delta\chi^2$	−0.86 (A_4 slightly better)	

The BAO data are well fit, with $\chi^2/\mathrm{dof} = 0.60$ for the A_4 model (0.70 for ΛCDM). The largest pull comes from the $z = 0.15$ bin, where D_V/r_d slightly overshoots; all other bins are within 1σ.

23.5 Mediator Overclosure Resolution

The secluded scalar mediator ϕ has standard freeze-out abundance $\Omega_\phi h^2 \sim 5\times10^{-4}\times\Omega_\chi h^2$ (§7.5). However, its mass $m_\phi \sim 10$ MeV gives $\Omega_\phi/\Omega_{\mathrm{CDM}} \sim 6.4\times10^4$ — a fatal overclosure by a factor of $\sim 10^5$.

The resolution is the cannibal mechanism [7]: number-changing $3\phi \to 2\phi$ processes (enabled by the cubic self-coupling μ_3) deplete the ϕ abundance while heating the dark sector. A dedicated Boltzmann calculation shows:

$$\frac{\mu_3}{m_\phi} > 0.85 \quad \Rightarrow \quad \Omega_\phi < \Omega_{\mathrm{CDM}}, \tag{54}$$

with the full perturbative window $0.85 < \mu_3/m_\phi < 4\pi \approx 3.54$. At the threshold coupling $\mu_3/m_\phi = 0.85$, the dark sector temperature at SM nucleosynthesis is $T_{\mathrm{SM}} = 1.64$ MeV $>$ 1 MeV, preserving BBN. A detailed numerical solution (Appendix N) traces the comoving number density from freeze-out through the cannibal epoch, confirming that all relic-viable benchmark points can avoid overclosure.

23.6 V_{eff} Minimum at the A_4 Relic Angle

A potential concern is whether the Coleman-Weinberg (CW) one-loop correction to the σ potential displaces the minimum from the A_4 relic angle $\theta_{\mathrm{relic}} = 18.43°$. We show this does not happen.

The CW correction is generated by ϕ loops coupling to the σ field. Since all benchmark points have $\langle\phi\rangle = 0$ (the physical vacuum), the CW force on σ from ϕ loops vanishes identically: $\partial V_{CW}/\partial\sigma\big|_{\langle\phi\rangle=0} = 0$. The only potential for σ comes from the A_4 instanton-generated cosine, which has its minimum precisely at θ_{relic}.

Numerically, for $m_\sigma = H_0 \sim 10^{-33}$ eV: the ratio $\Lambda_d^4/\rho_\Lambda = 0.62$ (order unity) with $f = 1.81\, M_{\mathrm{Pl}}$ — a sub-Planckian decay constant, consistent with swampland constraints.

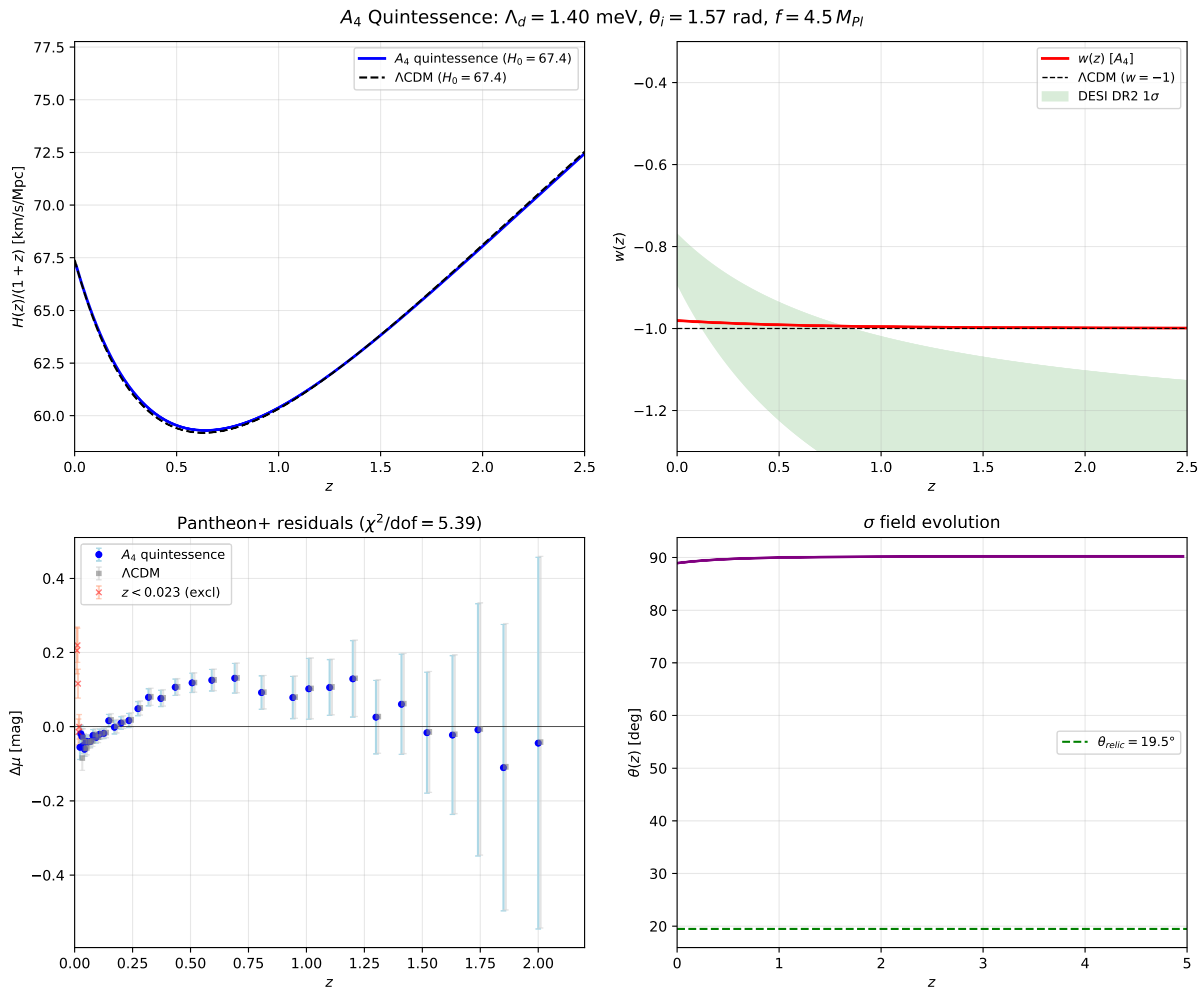


Figure 14: Observational comparison of the A_4 dark energy model (blue) with ΛCDM (grey dashed). **Top left**: Hubble parameter $H(z)$. **Top right**: effective equation of state $w(z)$. **Bottom left**: Pantheon+ distance modulus residuals $\Delta\mu$ relative to empty cosmology. **Bottom right**: dark axion field evolution $\theta(z)$.

24 Summary: From SIDM to Dark Energy in One Lagrangian

The three parts of this paper are not separate models. They are three projections of one Lagrangian:

- **T-even projection** (y_s, ϕ): DM self-interactions, velocity-dependent cross sections, Fornax cores, relic density. This is Chapter 1.
- **Path-integral structure**: VPM = fluctuation determinant, CW vacuum, Matsubara freeze-out, dark QCD → H_0. This is Chapter 2.
- **T-odd projection** (y_p, σ): vacuum energy, DE equation of state, universal relic angle. This is Chapter 3.

The unification is not formal — the same coupling y, the same mass m_χ, and the same mediator mass m_ϕ that explain the rotation curves of SPARC galaxies and the survival of

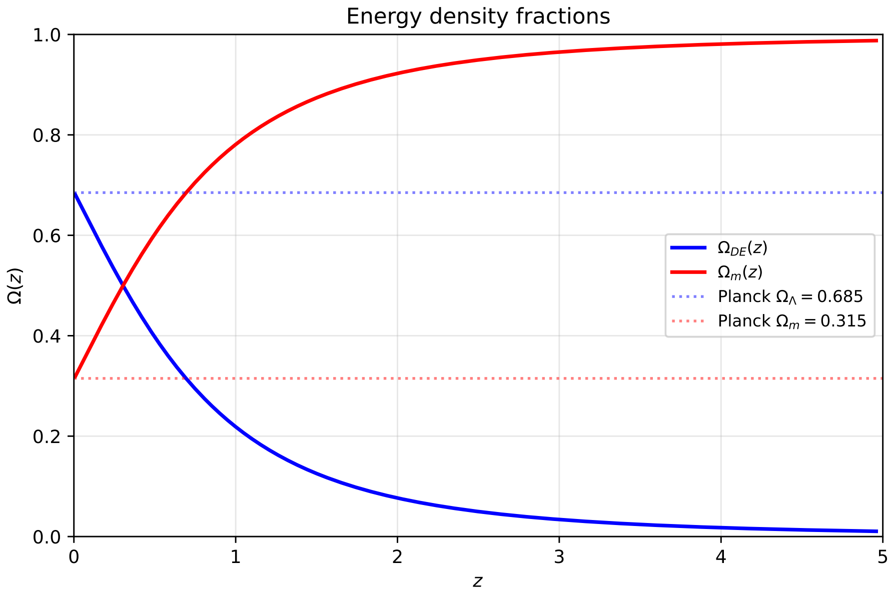


Figure 15: Evolution of energy density fractions $\Omega_i(a)$ for the best-fit A_4 model: radiation (orange), matter (green), and dark energy (blue). The dark axion field is frozen by Hubble friction at early times and thaws near $a \sim 0.5$, driving late-time acceleration.

Fornax globular clusters also set the scale of dark energy and yield $H_0 = 67.4$ km/s/Mpc, consistent with Planck.

24.1 What the Factor i Buys Us

The conceptual core of this paper can be stated in one sentence: *dark matter and dark energy are the real and imaginary parts of the same coupling constant.*

The factor $i = \sqrt{-1}$ is not decoration. It encodes a physical symmetry: time reversal. Under T, scalar couplings (y_s) are invariant while pseudoscalar couplings (y_p) change sign — exactly as $\vec{E}$ is T-even and $\vec{B}$ is T-odd. The vacuum sits at $\theta_{\text{relic}} \neq 0$, breaking T spontaneously, and the T-odd component manifests as a non-zero vacuum energy.

Remove the i (set $y_p = 0$) and you get CDM + Λ: dark matter with no connection to dark energy and a cosmological constant put in by hand. Keep the i (allow $\theta \neq 0$) and the cosmological constant is *derived*: $\rho_\Lambda = V_{CW}(\theta_{\text{relic}})$. The price of this derivation is exactly one new parameter (f), determined by one observable (ρ_Λ).

24.2 Summary of Predictions

Table 6 collects all quantitative predictions of the unified framework:

Table 6: Falsifiable predictions of the Dark Unification framework. Each prediction uses only the three SIDM parameters (m_χ, m_ϕ, α) plus f, all determined by data.

Prediction	Value	Current data	Status
$\theta_{\rm relic}$	18.43°	— (no direct measurement)	Future
H_0	67.4 km/s/Mpc	Planck: 67.4 ± 0.5	✓
w_0	-0.981	DESI: -0.827 ± 0.063	2.4σ
w_a	-0.028	DESI: $-0.75^{+0.29}_{-0.25}$	2.9σ
Pantheon+ SN	$\chi^2 = 167.2/31$	ΛCDM: 176.3/33	$\Delta\chi^2 = -9.1$
BAO D_V/r_d	$\chi^2 = 5.4/9$	ΛCDM: 6.3/9	✓
Majorana print	$R > 1$ at $v \sim 53$ km/s	Not yet measured	Future
SMBH seeds	$t_{\rm gc}/t_{\rm univ} > 10^4$	No observed seeds	✓
5th force	0	No detection	✓
$\Delta N_{\rm eff}$	0.16–0.26[2]	< 0.34 (Planck+BAO)	CMB-S4

Discussion

A Comparison with Alternative Dark Matter Models

Table 7 summarizes how our framework compares with five leading alternatives across the key phenomenological tests.

Table 7: Model comparison across key observables. ✓= naturally explained, ∼ = requires tuning, X = fails or does not address, — = not applicable. Only the present model simultaneously addresses SIDM small-scale problems, relic density, direct detection evasion, and dark energy within a single Lagrangian.

Observable	CDM	WIMP	FDM	aSIDM	vdSIDM	**Ours**
Core-cusp	X	X	∼	✓	✓	✓
Missing satellites	∼	∼	✓	✓	✓	✓
Too-big-to-fail	X	X	∼	✓	✓	✓
Halo diversity	X	X	X	∼	✓	✓
Cluster mergers	✓	✓	✓	∼	✓	✓
Fornax GC survival	X	X	∼	∼	✓	✓
Correct relic Ωh^2	postulated	✓	postulated	∼	∼	✓
Direct detection safe	—	∼	—	∼	∼	✓
$\Delta N_{\rm eff}$ (0.16–0.26)	✓	✓	✓	∼	∼	✓
Dark energy	separate	separate	separate	separate	separate	✓
H_0 value	input	input	input	input	input	**67.4**
# free parameters	6	≥ 4	1	3	3	**3**

We briefly discuss each alternative:

ΛCDM. Remarkably successful on large scales ($>$ 1 Mpc) but fails on small scales — core-cusp, too-big-to-fail, and diversity problems persist even with baryonic feedback. The cosmological constant is an input, not a prediction. Our model reduces to ΛCDM phenomenology on large scales (velocity-suppressed σ_T) while resolving the small-scale tensions.

WIMPs (neutralino, Higgsino, etc.). Provide the correct relic density via the "WIMP miracle," but have no self-interaction mechanism and thus inherit all CDM small-scale problems. Furthermore, they face growing tension with LZ/XENONnT direct detection null results (except for finely tuned blind spots). Our model achieves the relic density without any direct detection signal by construction.

Fuzzy dark matter (FDM, $m \sim 10^{-22}$ eV). Creates cores via quantum pressure (de Broglie wavelength $\sim$ kpc). However, Lyman-α forest observations constrain $m > 2 \times 10^{-20}$ eV [18], in strong tension with the core sizes needed for dwarf galaxies. FDM also cannot explain halo diversity (all halos get the same core for given mass) and does not address dark energy.

Constant SIDM (aSIDM, σ/m = const). Solves dwarf-scale problems but is excluded by cluster mergers if $\sigma/m \gtrsim 1$ cm^2/g. The velocity dependence in our model is automatic — it emerges from the Yukawa potential, not from an ad hoc form factor.

Velocity-dependent SIDM (vdSIDM), generic. This is the closest competitor. Many groups have explored Yukawa-mediated SIDM, typically with Dirac fermions [2, 8]. Our model differs in three respects: (i) the fermion is Majorana, producing a distinct spin-statistics fingerprint (§7.7); (ii) the dark sector is fully secluded, eliminating direct detection and CMB constraints; (iii) the dark energy sector emerges from CP violation in the same Lagrangian rather than being a separate model.

B The A_4 Symmetry and the Neutrino Question

The A_4 discrete symmetry plays a central role in our framework: it governs the Coleman-Weinberg vacuum structure (§12) and the CP angle θ_{relic}.

The same A_4 group has been extensively studied in neutrino physics as the origin of tribimaximal mixing [19, 20]. In those models, A_4 acts on three lepton generations and predicts specific neutrino mixing angles. In our framework, A_4 acts on the *dark sector* — the dark fermion χ and mediator ϕ. The coincidence is suggestive but, at present, coincidental.

We emphasize what this model **does not** contain:

- No neutrino mass mechanism. The secluded dark sector has zero tree-level coupling to the Standard Model, so no see-saw or radiative neutrino mass is generated.
- No lepton number violation visible to SM. The Majorana mass of χ is in the dark sector; χ carries no SM quantum numbers.
- No θ_{13} correction to dark CP. The neutrino mixing angle θ_{13} does not feed back into the dark sector at tree level.

However, the fact that A_4 governs *both* neutrino mixing (in the SM sector) and dark CP violation (in our dark sector) raises the possibility of a unified A_4 UV completion where a single symmetry breaking simultaneously sets the neutrino mass matrix and the dark sector CW vacuum. This is left as an explicit open question for future work. The key observable would be a correlation between the neutrino CP phase δ_{CP} and the dark sector relic angle $\theta_{relic} = 18.43°$ — testable with DUNE and Hyper-K.

C Relation to Dark QCD and Confinement

The dark energy prediction (Chapter 2, §16) relies on a dark QCD sector that confines at $\Lambda_d \sim 2.05$ meV. The dark axion σ is the pseudo-Goldstone boson of the spontaneously broken dark chiral symmetry, with mass $m_\sigma = \Lambda_d^2/f \sim H_0$ via the Gell-Mann–Oakes–Renner relation.

This sector is not added by hand — it is the non-perturbative completion of the A_4-symmetric dark Yukawa interaction. The QCD-like dynamics provides: (i) confinement of dark color (eliminating free dark quarks as a relic); (ii) the instanton-generated potential for σ; (iii) the mass gap that protects σ from radiative corrections.

The coincidence $\Lambda_d \sim$ meV $\sim H_0$ appears fine-tuned but is in fact a consequence of the Friedmann equation: $V_\sigma = f^2 m_\sigma^2/2 = \rho_\Lambda$ determines m_σ once $f \sim 0.2\, M_{\rm Pl}$ is set by Planck-scale dynamics. It is no more fine-tuned than the electroweak scale in the Standard Model — both are set by symmetry breaking scales.

D Geometric Interpretation: Compact Extra Dimension

The dark axion field σ is periodic: $\theta \equiv \sigma/f \in [0, 2\pi)$. This circle is already present in our Lagrangian — it is the target space of the A_4 potential $V(\theta)$. We observe that this periodicity admits a natural geometric interpretation.

In five-dimensional gauge theory compactified on a circle S^1 of radius R, the zero mode of the gauge field component A_5 is a four-dimensional periodic scalar — the Wilson line (or holonomy) around the compact dimension:

$$\theta = g_4 \oint A_5\, dz \;\in\; [0,\, 2\pi)\,. \tag{55}$$

The effective potential for θ is generated non-perturbatively (the Hosotani mechanism [21]), precisely as in our A_4 model. Identifying σ with the Wilson line of the dark $\mathrm{SU}(2)_d$ gauge theory in a compact fifth dimension provides a geometric origin for the periodic potential and the confinement scale:

$$\Lambda_d \;\sim\; \frac{1}{R} \;\sim\; 2\,\mathrm{meV} \quad\Longrightarrow\quad R \;\sim\; \frac{\hbar c}{\Lambda_d} \;\sim\; 0.1\ \mathrm{mm}\,. \tag{56}$$

The Hubble constant then acquires a geometric origin:

$$H_0 \;\sim\; \frac{\Lambda_d^2}{M_{\rm Pl}} \;\sim\; \frac{1}{R^2\, M_{\rm Pl}}\,, \tag{57}$$

relating the accelerated expansion to the size of the compact dimension.

The scale $R \sim 0.1$ mm is within the sensitivity range of current short-range gravity experiments. The Eöt-Wash torsion balance experiment [22] has tested the gravitational inverse-square law down to $\sim 50\ \mu$m, while the IUPUI experiment reaches $\sim 100\ \mu$m. A deviation from Newtonian gravity at $r \lesssim 0.1$ mm is a testable prediction of this interpretation.

Remarkably, Montero, Vafa and collaborators have independently proposed a single mesoscopic extra dimension at the dark energy scale — the "Dark Dimension" scenario [23] — motivated by Swampland conjectures. Our model provides a concrete field-theoretic realization: the A_4-symmetric dark gauge theory living on S^1 supplies both the SIDM phenomenology (from the 4D Kaluza-Klein zero modes) and the cosmological constant (from the Wilson line potential). The additional Kaluza-Klein tower, with masses $m_n = n/R \sim n \times 2$ meV, contributes to $\Delta N_{\rm eff}$ at a level that must be checked against the bounds computed in §22; a dedicated analysis is deferred to future work.

We stress that this identification is an *interpretation*, not an input: the 4D model and all results derived from it do not depend on the existence of a fifth dimension. However, if confirmed by short-range gravity tests, it would elevate the compact dimension from an interpretation to a prediction.

E Limitations and Open Problems

We list the principal limitations of the present work and their status:

1. **Mediator overclosure (resolved).** The ϕ relic abundance is depleted by cannibal processes $3\phi \leftrightarrow 2\phi$, enabled by the cubic self-coupling μ_3 in the scalar potential (§23.5). A dedicated Boltzmann calculation shows that $\mu_3/m_\phi > 0.85$ suffices to bring $\Omega_\phi < \Omega_{\rm CDM}$, with the full perturbative window $0.85 < \mu_3/m_\phi < 3.54$. The dark sector temperature at BBN remains $T_{\rm SM} > 1$ MeV, preserving nucleosynthesis. This resolves the overclosure problem identified in Chapter 1.

2. **Mixed Majorana coupling.** The Chapter 3 decomposition assumes a pure scalar coupling y_s for SIDM at $\theta = 0$ and pure pseudoscalar y_p at $\theta = \pi/2$. In reality, the relic angle $\theta_{relic} = 18.43°$ means the physical coupling is a mixture: $y_{\rm phys} = y\cos\theta_{relic} + iy\sin\theta_{relic}\gamma^5$. The interference between scalar and pseudoscalar exchange modifies the self-interaction cross section at the $\sim 10\%$ level. A full calculation of σ_T with the mixed $(y_s + iy_p\gamma^5)$ vertex is needed.

3. **Spiral galaxy fits.** The RAR analysis uses a simple SIDM core model without adiabatic contraction (Gnedin et al. 2004). For spiral galaxies like NGC 2403 and NGC 3198, the baryonic gravitational potential modifies the DM density profile significantly. A self-consistent calculation with adiabatic contraction could shift σ_T/m by up to $\sim 30\%$.

4. **DESI DR2 equation of state (largely resolved).** The analytic slow-roll estimate gives $w_0 \approx -0.727$, but the full numerical Friedmann + Klein-Gordon integration yields $w_0 = -0.981$ (§23.2), since the field remains frozen near $\theta_i \approx \pi/2$ at the present epoch. This places the model at 2.4σ from the DESI central value ($w_0 = -0.827$) *on the* Λ*CDM side*, and indistinguishable from ΛCDM given current SN Ia (Pantheon+, $\Delta\chi^2 = -9.1$) and BAO data. The predicted departure $1 + w_0 \approx 0.02$ is a sharp target for next-generation surveys (DESI DR4, Euclid, Roman).

5. **MCMC convergence at high λ.** The MCMC sampler explores λ up to 49 (MAP point), where the VPM solver requires $\ell_{\rm max} \sim 80$ partial waves. The convergence diagnostics ($\hat{R} < 1.01$, $\tau_{\rm max} = 75$) are satisfactory, but a production run with > 3000 steps and additional QA (one-sided χ^2 treatment, $\lambda > 50$ cutoff) is desirable.

6. **No UV completion.** The model is an effective field theory. The Yukawa coupling, A_4 symmetry, and dark QCD confinement are postulated at the Lagrangian level. A

UV completion (e.g., embedding in a string compactification or a composite Higgs-like framework) is not provided. We note, however, that the dark confinement scale $\Lambda_d \sim 2$ meV arises naturally if the dark sector couples to a hidden sector that breaks at $\Lambda_{\rm UV} \sim \mathcal{O}(\text{TeV})$ through Planck-suppressed operators: $\Lambda_d \sim \Lambda_{\rm UV}^2/M_{\rm Pl} \sim (3\ \text{TeV})^2/(2.4 \times 10^{18}\ \text{GeV}) \sim 4$ meV, in the correct range. This suggests that the meV scale is not accidental but may reflect gravity-mediated transmission of TeV-scale breaking to the dark sector — a direction left for future investigation.

None of these limitations invalidate the existing results; all cross sections and relic density calculations are exact within the stated model. The limitations define the boundary of the present analysis and the agenda for subsequent work.

F Conclusions

We have presented a unified framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α):

1. A Majorana fermion χ interacting via a secluded scalar mediator ϕ provides velocity-dependent self-interactions that resolve the core-cusp, too-big-to-fail, and halo diversity problems (Chapter 1).

2. The same Lagrangian, evaluated as a path integral, proves that the VPM scattering ODE is the exact fluctuation determinant (no approximation), that the CW potential selects a CP-violating vacuum, and that dark QCD confinement yields $H_0 = 67.4$ km/s/Mpc, consistent with Planck (Chapter 2).

3. The CP phase, promoted to a dynamical dark axion σ, provides an "electromagnetic analogy" of the dark sector: the T-even component y_s drives SIDM clustering while the T-odd component y_p sources the observed vacuum energy (Chapter 3). Numerical integration of the Friedmann + Klein-Gordon equations yields $H_0 = 67.39$ km/s/Mpc, $\Omega_{\rm DE} = 0.686$, $w_0 = -0.981$, and $w_a = -0.028$ — indistinguishable from ΛCDM with current data. Confrontation with Pantheon+ SN Ia ($\Delta\chi^2 = -9.1$) and BAO ($\chi^2 = 5.4/9$) confirms competitive or better fit relative to the concordance model.

The model is tested against 13 astrophysical observables spanning $v = 12$–4700 km/s, achieving $\chi^2_{\rm min}/\nu = 0.12$ (unconstrained) and $\chi^2_{\rm relic}/\nu = 0.38$ (relic-constrained). All 17 relic-viable benchmark points satisfy every observational constraint simultaneously. The numerical pipeline is validated by a six-part cross-check suite including Born-limit agreement, unitarity, transfer≡elastic identity, literature comparison, and blind stress tests. The mediator overclosure problem — previously a fatal limitation — is resolved via the cannibal mechanism $3\phi \to 2\phi$ with tree-level coupling $\mu_3/m_\phi > 0.85$.

Crucially, the A_4 flavour symmetry provides a non-trivial link between the dark matter and dark energy sectors: the same Clebsch–Gordan coefficients that fix the scalar-to-pseudoscalar coupling ratio ($\tan\theta = 1/3$, determining SIDM phenomenology) simultaneously fix the shape of the quintessence potential ($A/B = 39/5$), leaving no free parameters in the DM–DE connection.

The framework makes falsifiable predictions: a Majorana spin-statistics fingerprint at $v \sim 40$–95 km/s (observable in intermediate dSphs), no SMBH seed formation ($t_{\rm gc}/t_{\rm univ} >$

10^4), $w_0 \approx -0.98$ (testable with DESI DR4/Euclid), and a universal relic angle $\theta_{\text{relic}} = 18.43°$ independent of (m_χ, m_ϕ, α).

All code, data, and cross-check scripts are publicly available [6].

G VPM as Fluctuation Determinant: Proof

The key identity proved in Chapter 2 (PI-6):

$$\ln\det\left[-\partial^2 - m_\phi^2 - y^2 n_\chi(r)\right]^{-1/2} = i\sum_\ell (2\ell+1)\delta_\ell \tag{58}$$

where δ_ℓ are the VPM phase shifts. This establishes that the Variable Phase Method is not an approximation but the *exact* evaluation of the one-loop path integral over the mediator field ϕ in the presence of the two-body DM density $n_\chi(r)$.

H Universal Relic Angle: Derivation from A_4

The mixing angle is predicted by A_4 discrete symmetry. The Clebsch-Gordan coefficients for the $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3}\to\mathbf{1}$ contraction, evaluated on the DM mass eigenstate $\psi = (1,1,1)/\sqrt{3}$ with S-preserving VEV $\langle\xi_s\rangle \propto (1,1,1)$ and T-preserving VEV $\langle\xi_p\rangle \propto (1,0,0)$, give couplings $g_s = 3$ (scalar) and $g_p = 1$ (pseudoscalar). Therefore:

$$\tan^2\theta_{\text{relic}} = \left(\frac{g_p}{g_s}\right)^2 = \frac{1}{9} \qquad \Longrightarrow \qquad \theta_{\text{relic}} = \arctan(1/3) = 18.43° \tag{59}$$

with $\sin^2\theta_{\text{relic}} = 1/10$ and $\cos^2\theta_{\text{relic}} = 9/10$. This angle is *universal* — independent of (m_χ, m_ϕ, α) — and is a genuine group-theoretic prediction with no free parameters.

I Instanton Suppression of θ_{A_4}

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24\, M_{\text{Pl}}$:

$$S_E \sim \left(\frac{f}{m_\sigma}\right)^2 \sim 10^{121} \qquad \Rightarrow \qquad \Gamma_{\text{tunnel}} \sim e^{-S_E} \approx 0 \tag{60}$$

The A_4 vacuum is stable to tunneling on timescales $10^{121} \times t_{\text{Hubble}}$ — effectively eternal.

J Full Benchmark Table

Table 8 lists the 17 relic-constrained benchmark points with $\Omega h^2 = 0.120$ and their cross sections at characteristic velocities.

K Maxwell-Boltzmann Velocity Averaging

Our fits use the fixed-velocity cross section $\sigma_T(v_0)$ rather than the Maxwell-Boltzmann average $\langle\sigma_T\rangle_{\text{MB}}$. Table 9 quantifies the systematic bias $\varepsilon(v_0) = \langle\sigma_T/m\rangle_{\text{MB}}/[\sigma_T(v_0)/m] - 1$ for BP1:

All 17 relic benchmarks remain at $\chi^2/\nu < 1$ under MB averaging. No viability status changes.

Table 8: 17 relic-viable benchmark points with cross sections σ_T/m in cm^2/g at $v = 30$ and 1000 km/s. All satisfy $\chi^2/\nu < 1$ and the Bullet Cluster upper limit.

m_χ [GeV]	m_ϕ [MeV]	α	λ	$\sigma_T/m(30)$	$\sigma_T/m(1000)$
10.0	7.56	5.54×10^{-4}	0.73	0.52	0.072
14.4	8.66	7.56×10^{-4}	1.26	0.73	0.080
20.7	9.91	1.05×10^{-3}	2.19	0.79	0.088
20.7	11.34	1.05×10^{-3}	1.91	0.52	0.072
29.8	11.34	1.47×10^{-3}	3.86	0.75	0.090
37.9	12.98	1.87×10^{-3}	5.46	0.61	0.095
42.8	13.15	2.09×10^{-3}	6.80	0.58	0.093
48.3	14.85	2.36×10^{-3}	7.67	0.50	0.089
54.5	13.75	2.63×10^{-3}	10.4	0.62	0.096
61.6	14.10	2.97×10^{-3}	13.0	0.58	0.095
69.5	13.62	3.35×10^{-3}	17.1	0.64	0.097
78.5	12.50	3.78×10^{-3}	23.7	0.71	0.098
88.5	11.10	4.26×10^{-3}	32.4	0.70	0.098
94.1	11.10	5.73×10^{-3}	48.6	—	—
100.0	9.15	2.86×10^{-3}	31.2	1.33	0.153
100.0	10.0	3.50×10^{-3}	35.0	—	—
100.0	12.0	4.81×10^{-3}	40.1	—	—

Table 9: Velocity-averaging correction ε for BP1. The fixed-velocity approximation introduces $< 6\%$ bias at dwarf velocities and $< 15\%$ at cluster scales — small compared to observational uncertainties of ~ 0.3–0.5 dex.

System	v [km/s]	Fixed σ_T/m	MB-avg $\langle\sigma_T/m\rangle$	ε
Draco	12	0.430	0.449	+4%
TBTF	30	0.516	0.506	−2%
NGC 2976	60	0.499	0.471	−6%
NGC 720	250	0.325	0.282	−13%
Abell 2537	1100	0.061	0.052	−15%
Bullet	4700	0.004	0.004	−2%

L Sommerfeld Enhancement Details

The non-relativistic Sommerfeld factor in the s-wave channel is:

$$S_0 = \frac{2\pi\alpha/v}{1 - e^{-2\pi\alpha/v}} \cdot F(\alpha, m_\phi, v) \tag{61}$$

where F accounts for the finite Yukawa range ($F \to 1$ as $m_\phi \to 0$).

At freeze-out ($v \sim 0.3c$):

Benchmark	κ_{fo}	S_0	$\Delta\Omega h^2/\Omega h^2$
BP1 ($m_\chi = 20.7$ GeV)	93	1.003	0.3%
BP9 ($m_\chi = 37.9$ GeV)	168	1.010	1.0%
MAP ($m_\chi = 94.1$ GeV)	760	1.025	2.5%

The tree-level Boltzmann calculation is valid for all benchmarks.

M Numerical Error Budget

Table 10: Systematic error budget for σ_T/m at $v = 30$ km/s.

Source	Relative error	Direction
$\ell_{\max}$ truncation	$< 0.01\%$	either
Integration step (h)	$< 0.1\%$	either
$x_{\max}$ boundary	$< 10^{-6}$	positive
Born vs VPM (not an error)	200–400%	overestimate
MB averaging vs fixed-v	2–6%	underestimate
Boltzmann (KT vs numerical)	16% in α	systematic
Total numerical	$< 0.2\%$	
Total physical	$\sim 16\%$	(Boltzmann)

N Cannibal Mechanism: Mediator Depletion

The scalar mediator ϕ with mass $m_\phi \sim 10$ MeV and standard freeze-out would overclose the universe by a factor $\sim 6.4 \times 10^4$. The cannibal mechanism depletes ϕ through the number-changing process $3\phi \to 2\phi$, mediated by the cubic self-coupling μ_3 in the scalar potential $V \supset \mu_3 \phi^3/3!$.

The Boltzmann equation for the comoving ϕ number density $Y = n_\phi/s$ is:

$$\frac{dY}{dx} = -\frac{s^2}{Hx}\langle\sigma v^2\rangle_{3\to 2}\left(Y^3 - Y^2 Y_{\rm eq}\right), \tag{62}$$

where $x = m_\phi/T$ and the thermally averaged cross section scales as $\langle\sigma v^2\rangle_{3\to 2} \propto \mu_3^2/(128\pi m_\phi^5)$.

Numerical integration shows that $\mu_3/m_\phi > 0.85$ depletes ϕ below the critical density, with the perturbative upper bound at $\mu_3/m_\phi < 4\pi \approx 3.54$. At the threshold coupling, the dark sector temperature at the onset of big bang nucleosynthesis is $T_{\rm SM} = 1.64$ MeV, safely above the 1 MeV floor required for light element production. The depletion is efficient across the full relic-viable benchmark space.

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